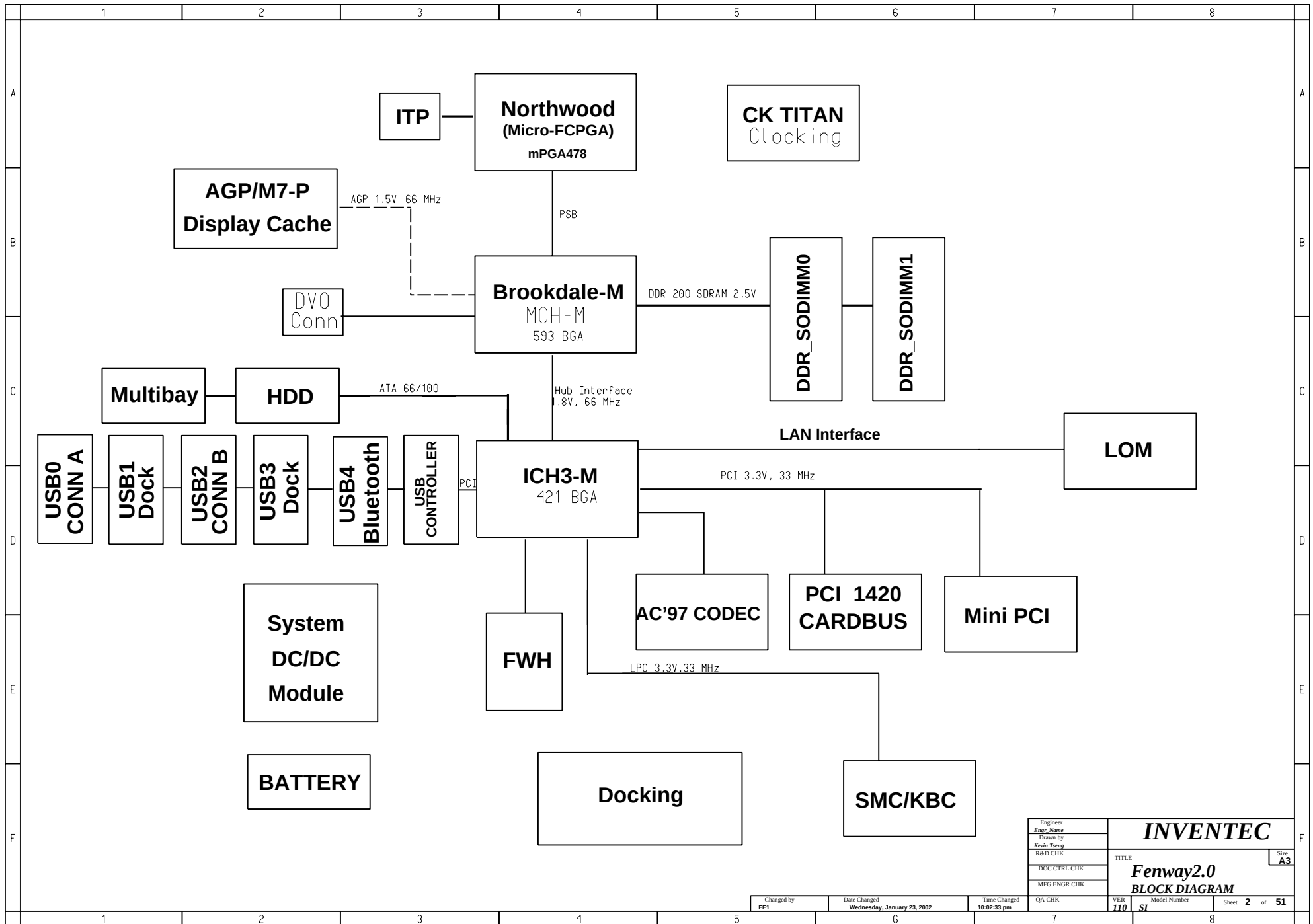
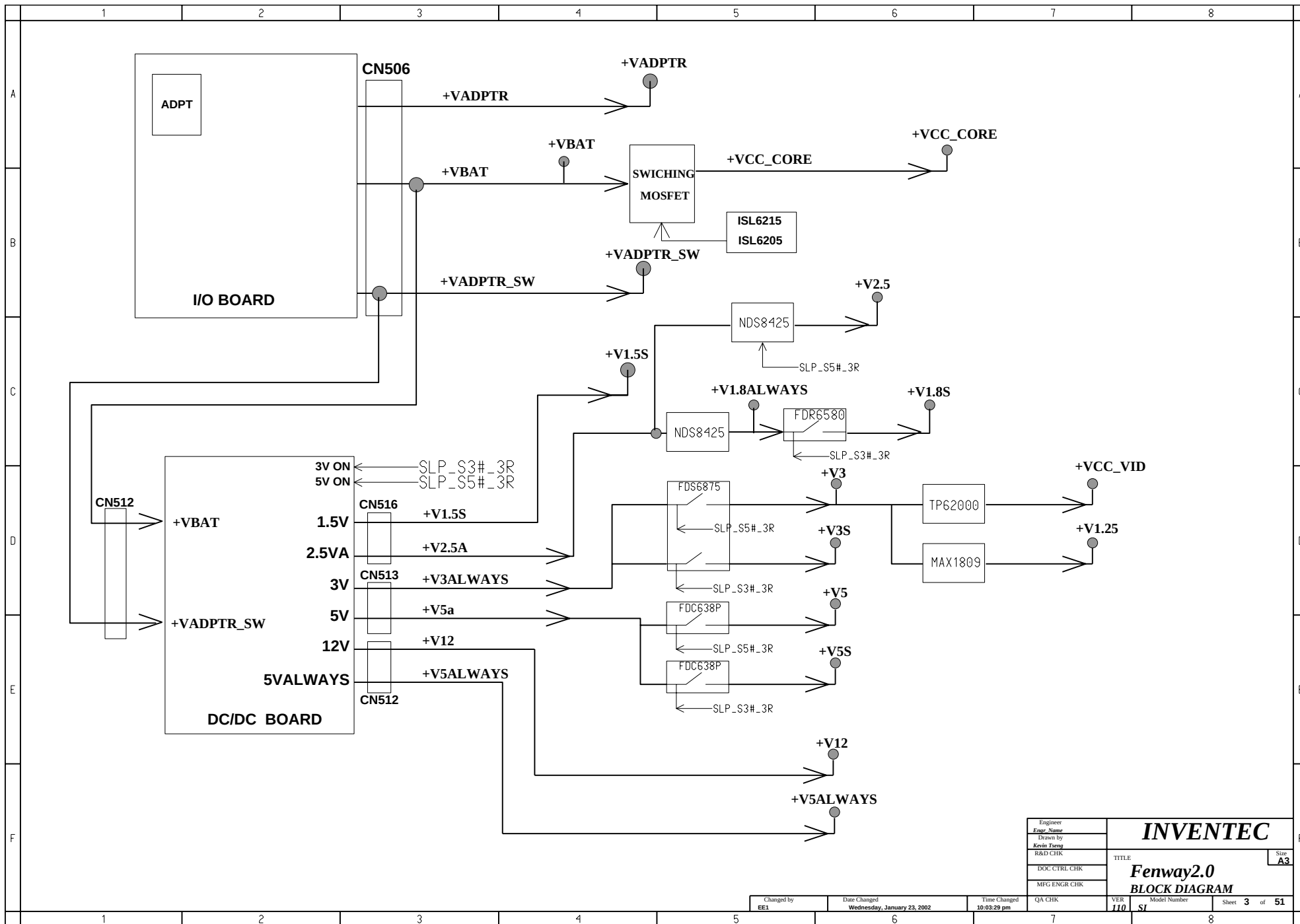


	1	2	3	4	5	6	7	8										
A	TABLE OF CONTENTS				Revisions		Approvals											
					REV	ECN NO	Description Of Change	DATE	DFTG.	ENGR.	REL.							
B	PAGE				PAGE													
	4 - CLOCK GENERATOR				32 - DUALBAY													
	5-7 - NORTHWOOD PROCESSOR				33 - INT KBD./ POINTING DEVICE													
	8-10 - BROOKDALE				34 - LID SWITCH/LEDS													
C	11-14 - DDR				35 - DOCKING CONNCTOR													
	15,16 - ATI_M7_P				36 - MINI PCI													
	17 - VIDEO RAM				37 - USB, IR													
	18 - CRT,LCD,INVERTER CONNECTOR INTERFACE				38 - BLUETOOTH													
D	19,20 - ICH3-M				39 - THERMAL SENSOR & FAN CONTROLLER													
	21 - KAHUNA				40 - ADAPTER SENSE CIRCUIT & CLAMP CIRCUIT													
	22 - FWH & EEPROM				41 - PWR MODULE CONN													
	23 - SERIAL , PARALLEL AND PS2 PORTS				42 - DECOUPLING CAPS													
E	24 - CARDBUS CONTROLLER				43 - DRILL HOLE													
	25 - PC CARD SLOT				44 - HIGH SPEED													
	26 - AC97 CODEC				45 - PULL UPS													
	27 - EQ & MIC JACK				46 - LAN INTERFACE													
F	28 - AUDIO AMP & HP JACK				47-49 - POWER													
	29 - HDD AND MB1 PWR AND SWITCH				50 - USB2.0													
	30 - HDD CONNECTOR				51 - IO BOARD													
	31 - MULTIBAY																	
					Changed by EE1		Date Changed Friday, April 19, 2002		Time Changed 9:58:54 am		QA CHK		VER 110		Model Number SI		Sheet 1 of 51	
	1	2	3	4	5	6	7	8										

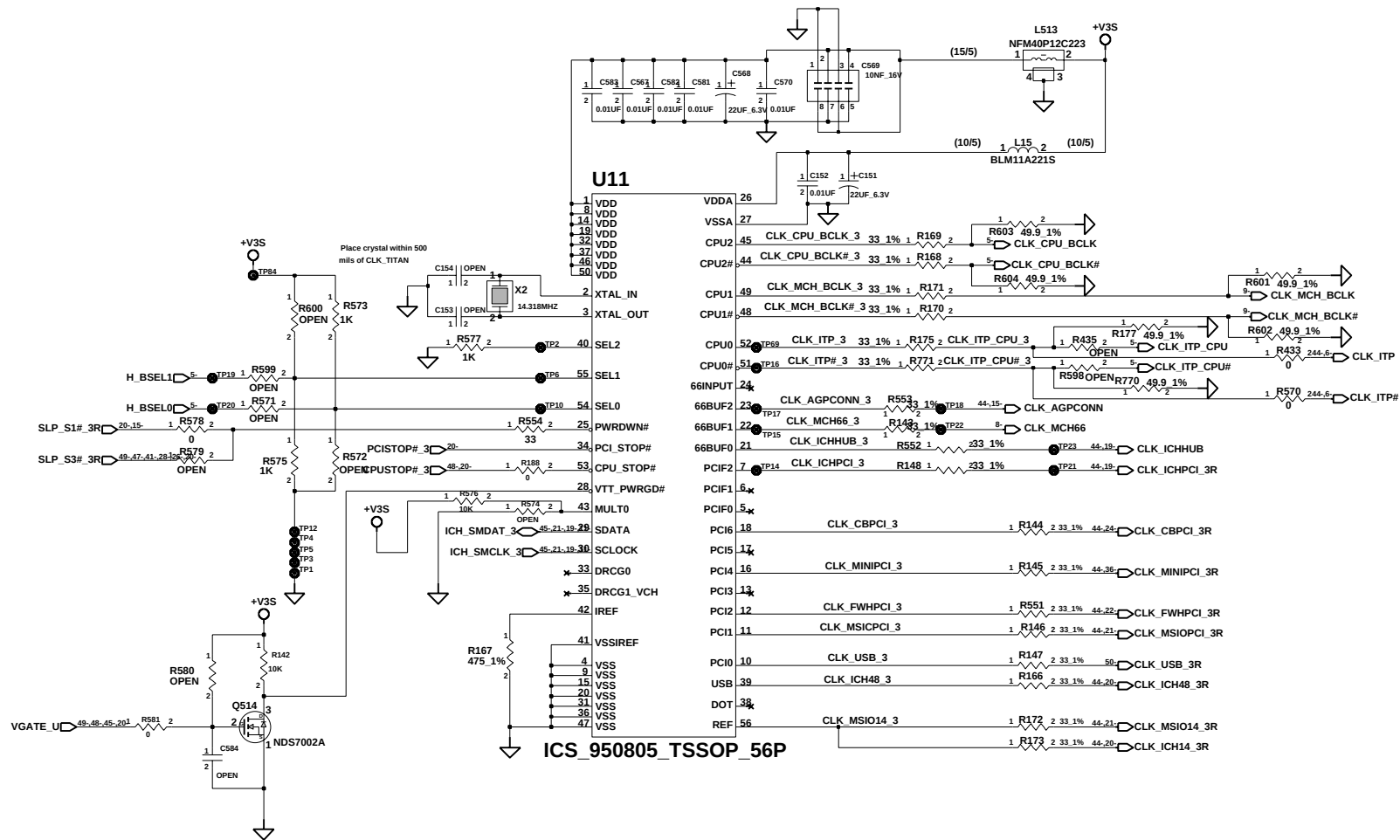
Engineer <i>Engy Name</i>	INVENTEC TITLE: Fenway2.0 TABLE OF CONTENTS		
Drawn by <i>Kevin Torng</i>			
R&D CHK			
DOC CTRL CHK			
MFG ENGR CHK	VER 110	Model Number SI	Sheet 1 of 51

Changed by EE1	Date Changed Friday, April 19, 2002	Time Changed 9:58:54 am
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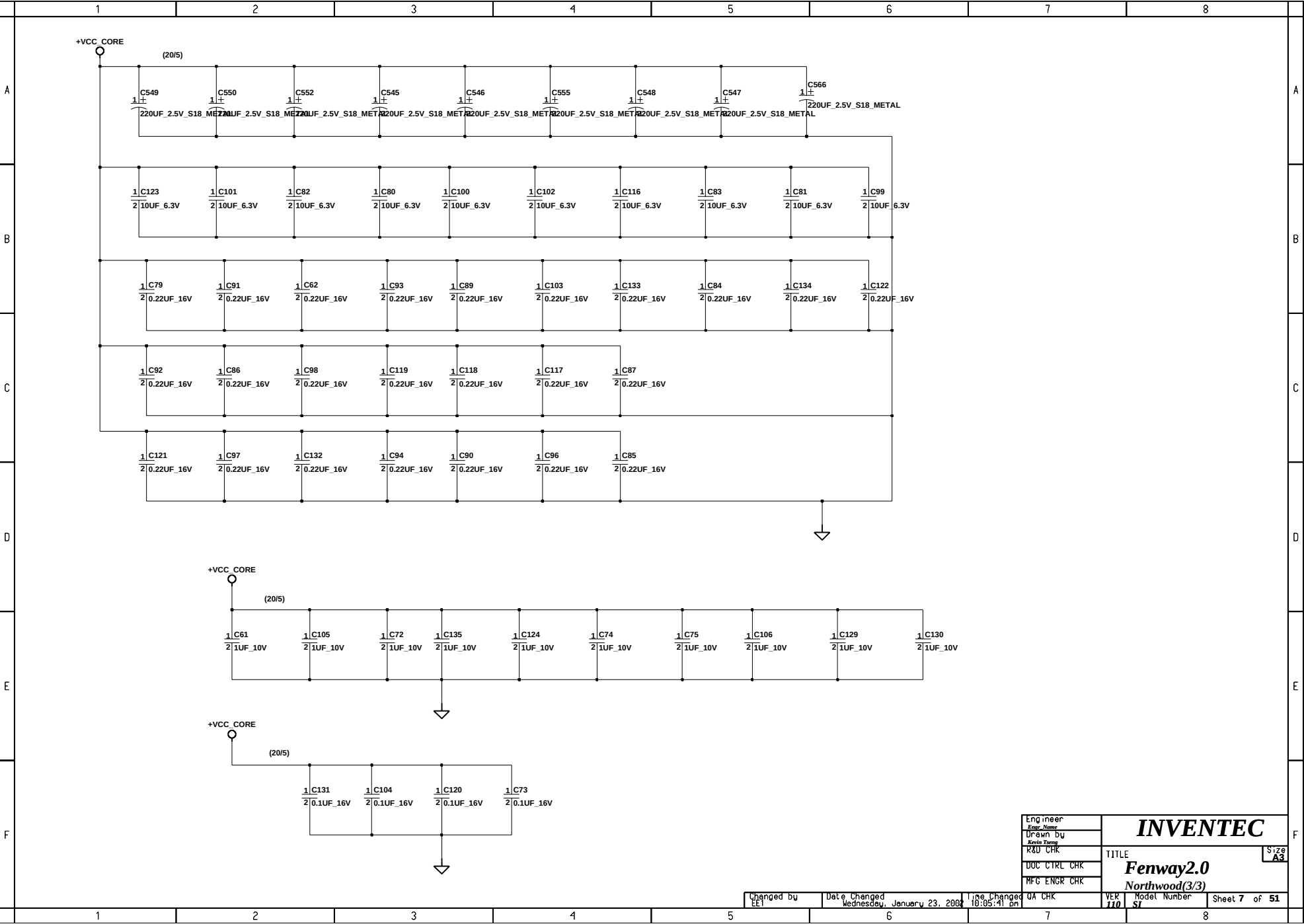


Engineer <i>Engr_Name</i>	INVENTEC TITLE Fenway2.0 BLOCK DIAGRAM Size A			
Drawn by <i>Kevin Tseng</i>				
R&D CHK				
DOC CTRL CHK				
MFG ENGR CHK				
QA CHK	VER <i>110</i>	Model Number <i>S1</i>	Sheet 3	of 51



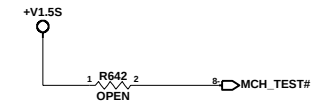
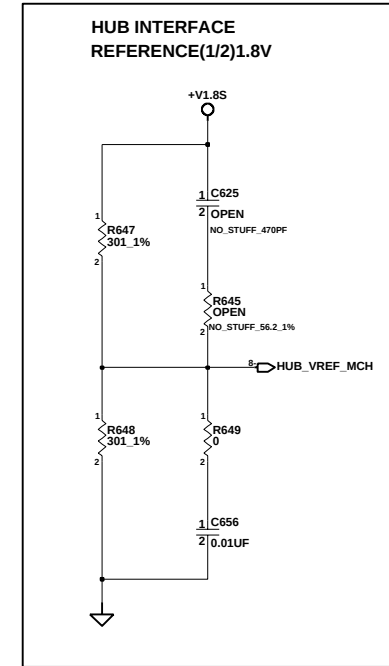
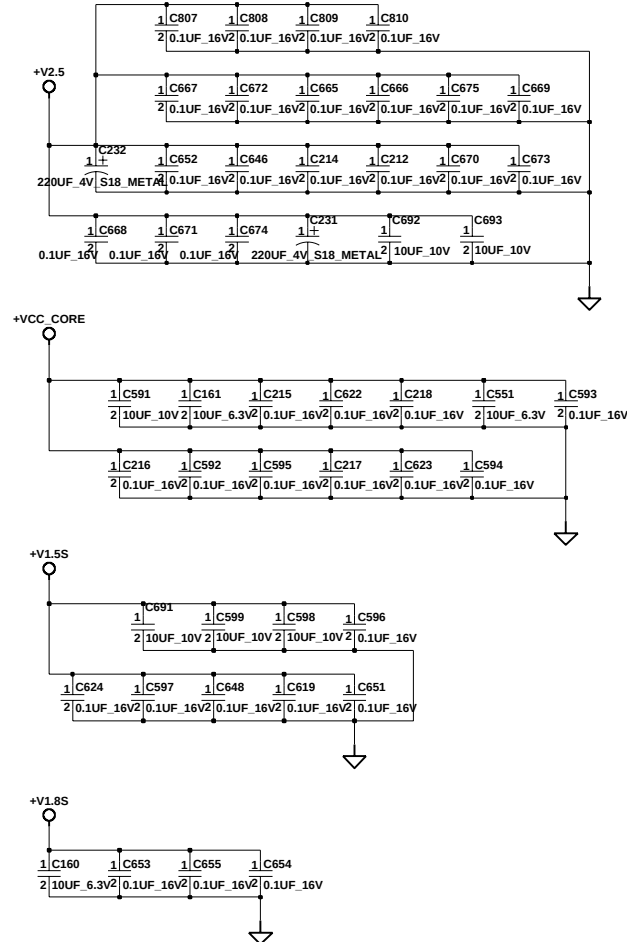
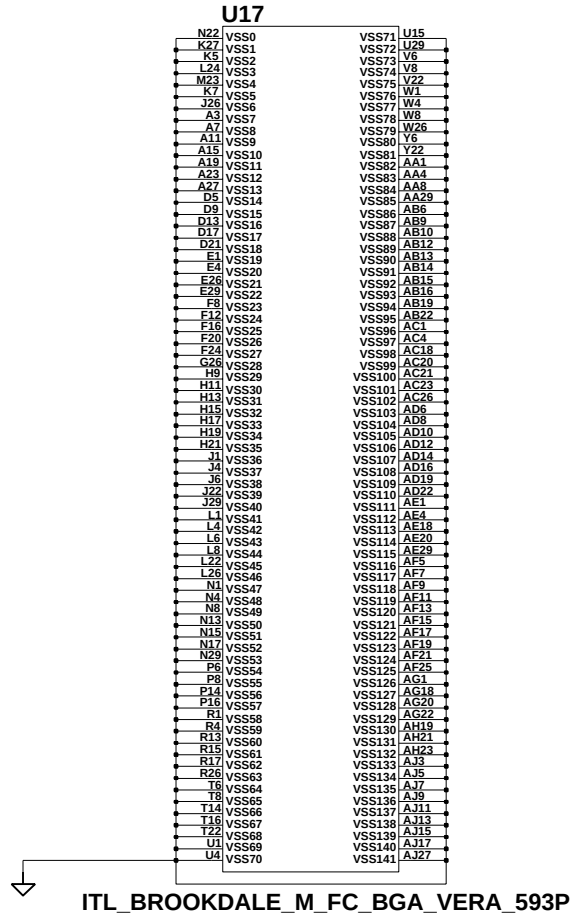
Engineer	INVENTEC	
Drawn by		
Rev'd	TITLE	
Rev'd		
DOC CTRL	Fenway2.0	
MFG ENGR		
Model Number	CLK TITAN	
VER	110	Sheet 4 of 51

Changed by	Date Changed	Time Changed
EE1	Tuesday, April 16, 2002	1:33:57 pm



Engineer	INVENTEC		
Drawn by			
Rev'd			
Doc. Ctrl. Chk			
Mfg. Engr. Chk			
TITLE		Northwood(3/3)	
Model Number		Sheet 7 of 51	
VER		110	

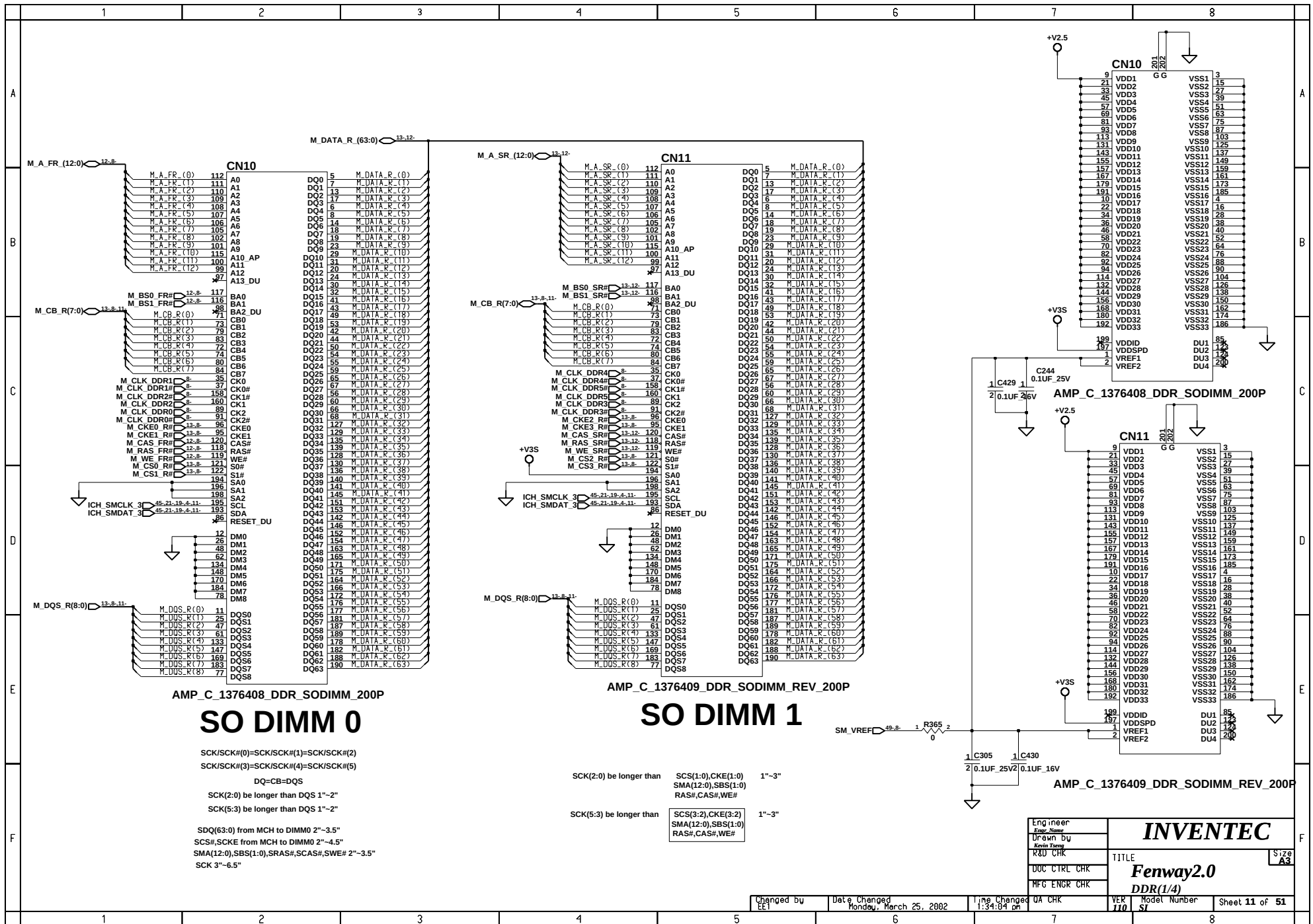
Changed by	Date Changed	Time Changed
EE1	Wednesday, January 23, 2008	10:05:41 am

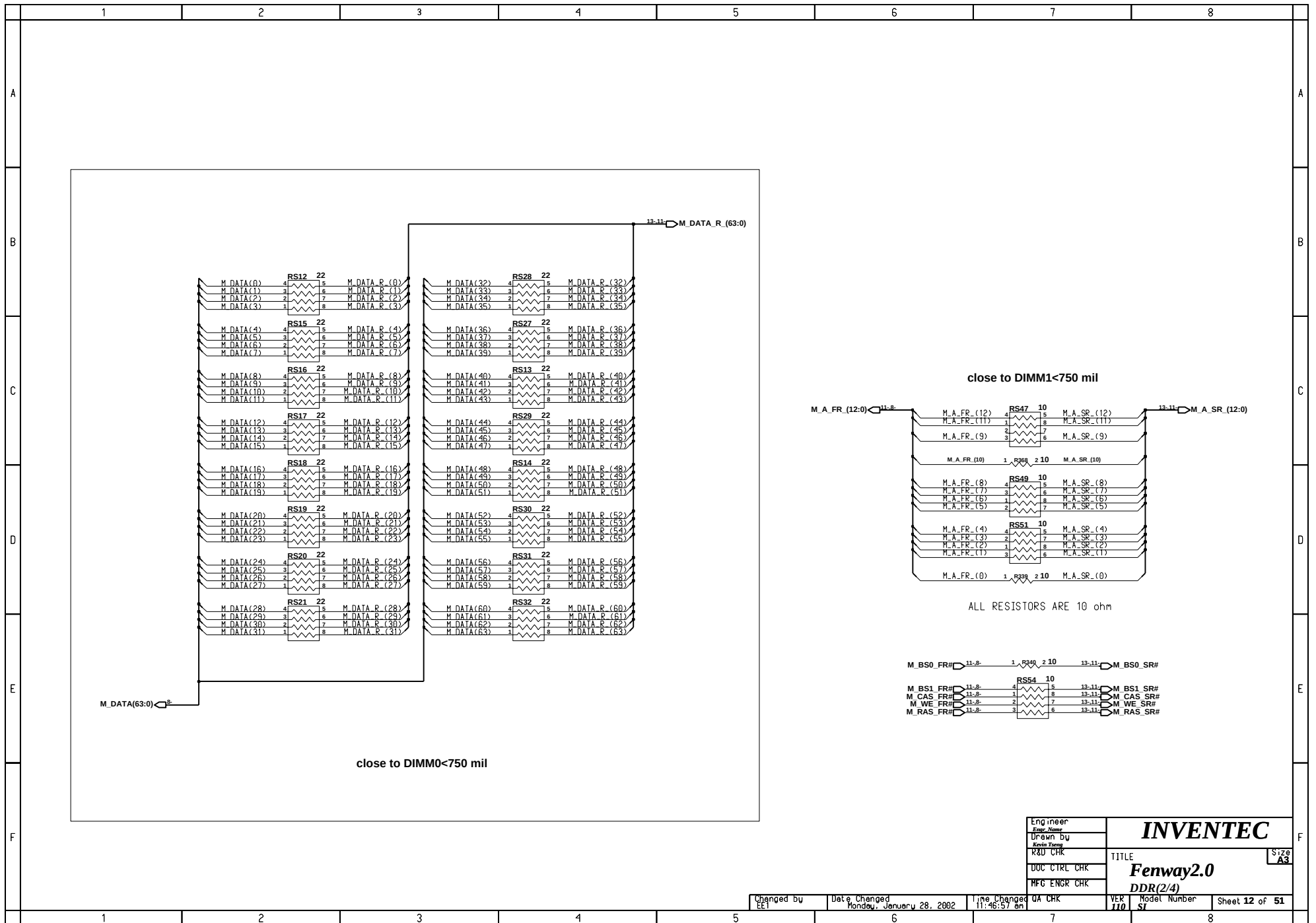


BROOKDALE 3 OF 3

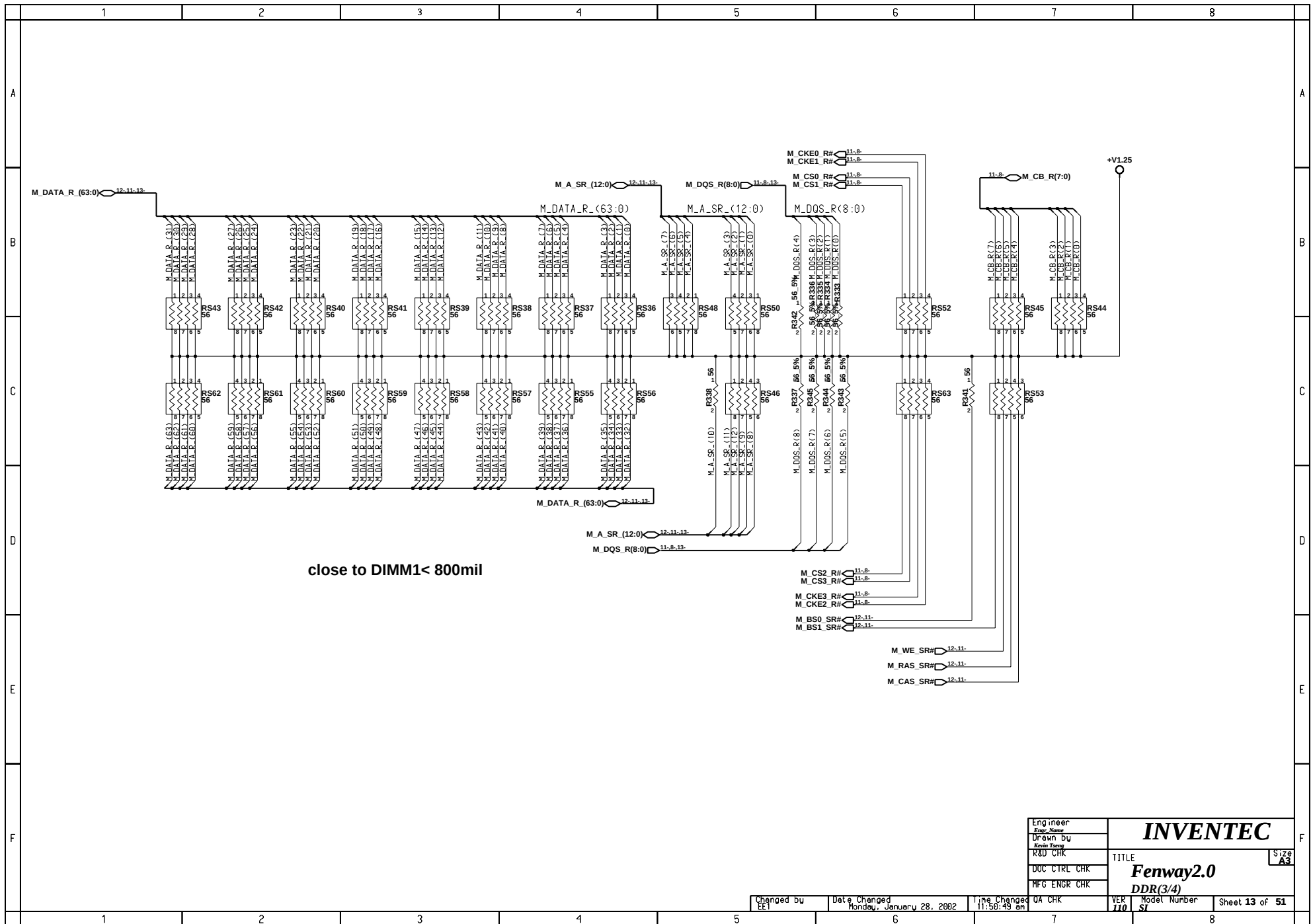
Engineer Eng Name	INVENTEC		
Drawn by Kevin Tarnag			
Rev'd CHK	TITLE		S:729 A3
DOC CTRL CHK	Fenway2.0		
MFG ENGR CHK	Brookdale(3/3)		
UA CHK	VER 110	Model Number 51	Sheet 10 of 51

Changed by EE1	Date Changed Friday, May 10, 2002	Time Changed 6:30:10 pm
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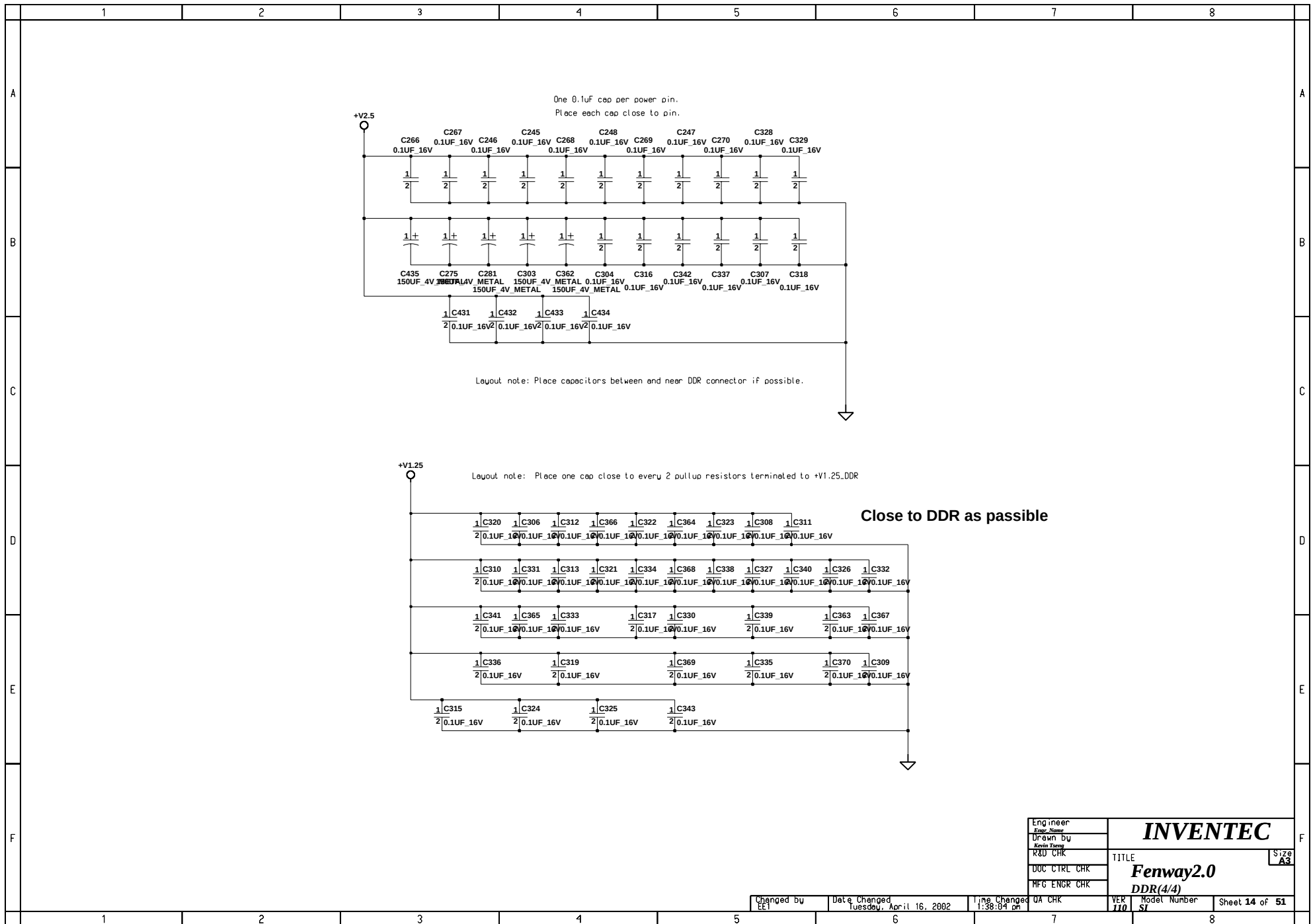


Engineer <i>Engin Name</i>	INVENTEC	
Drawn by <i>Kevin Tzeng</i>		
R&D CHK	TITLE	S/78 A3
DOC CTRL CHK	Fenway2.0	
MFG ENGR CHK	DDR(2/4)	
UA CHK	VER 110	Model Number 51
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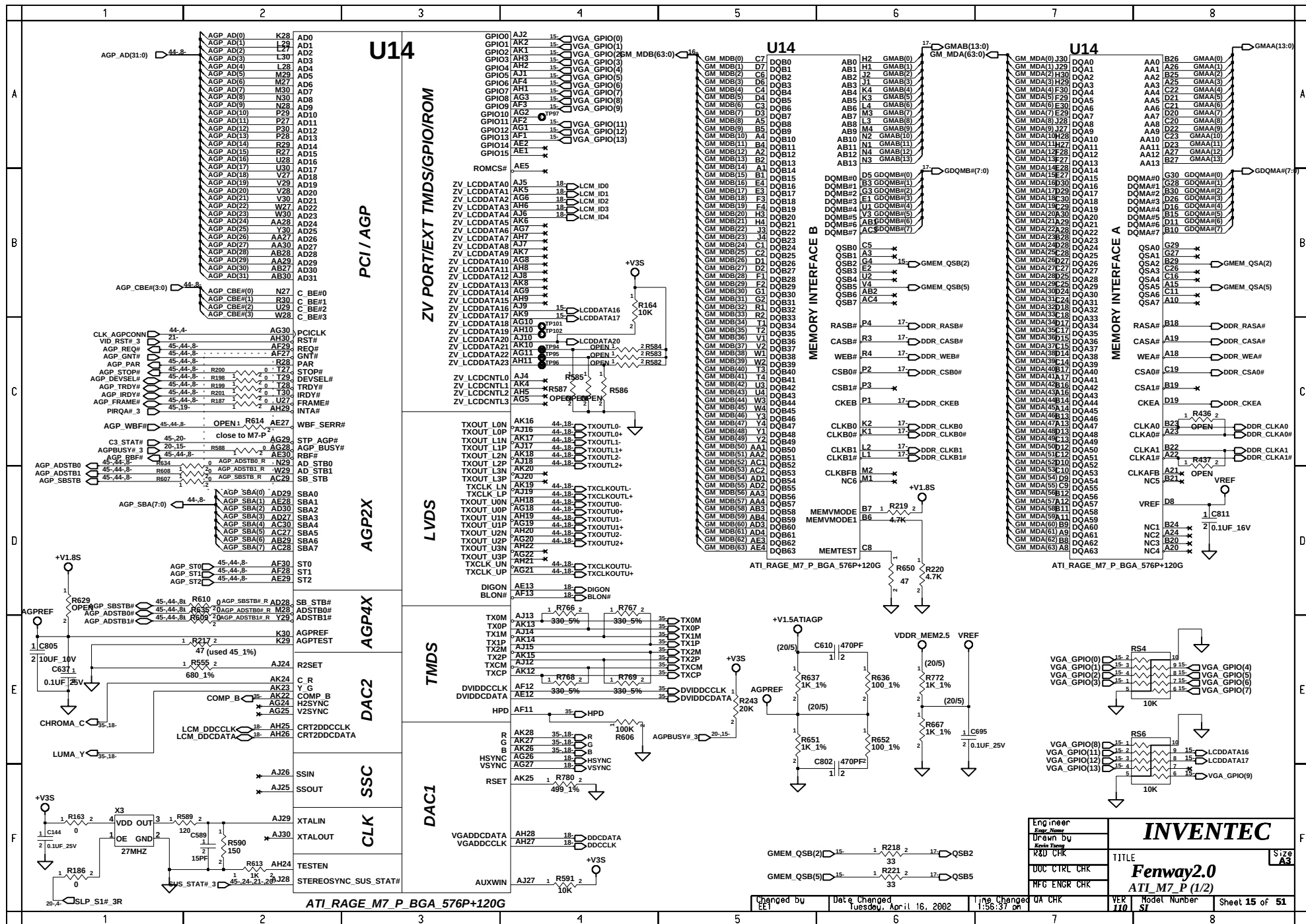


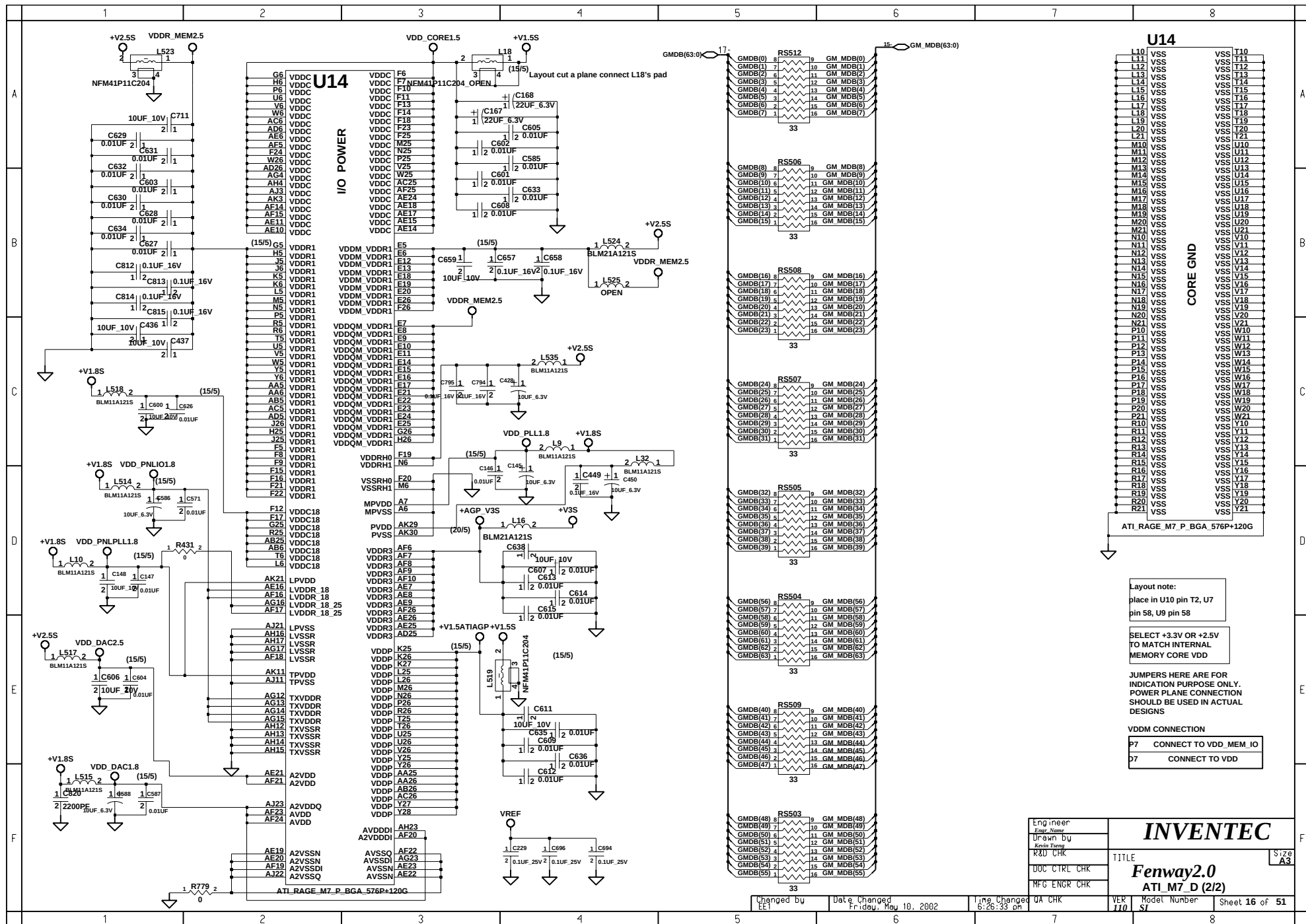
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Drawn by			
Rev'd	S:78 A3		
Rev'd			
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PRG ENGR CHK			
QA CHK	Model Number DDR(3/4)		
VER 110			
Sheet 13 of 51			

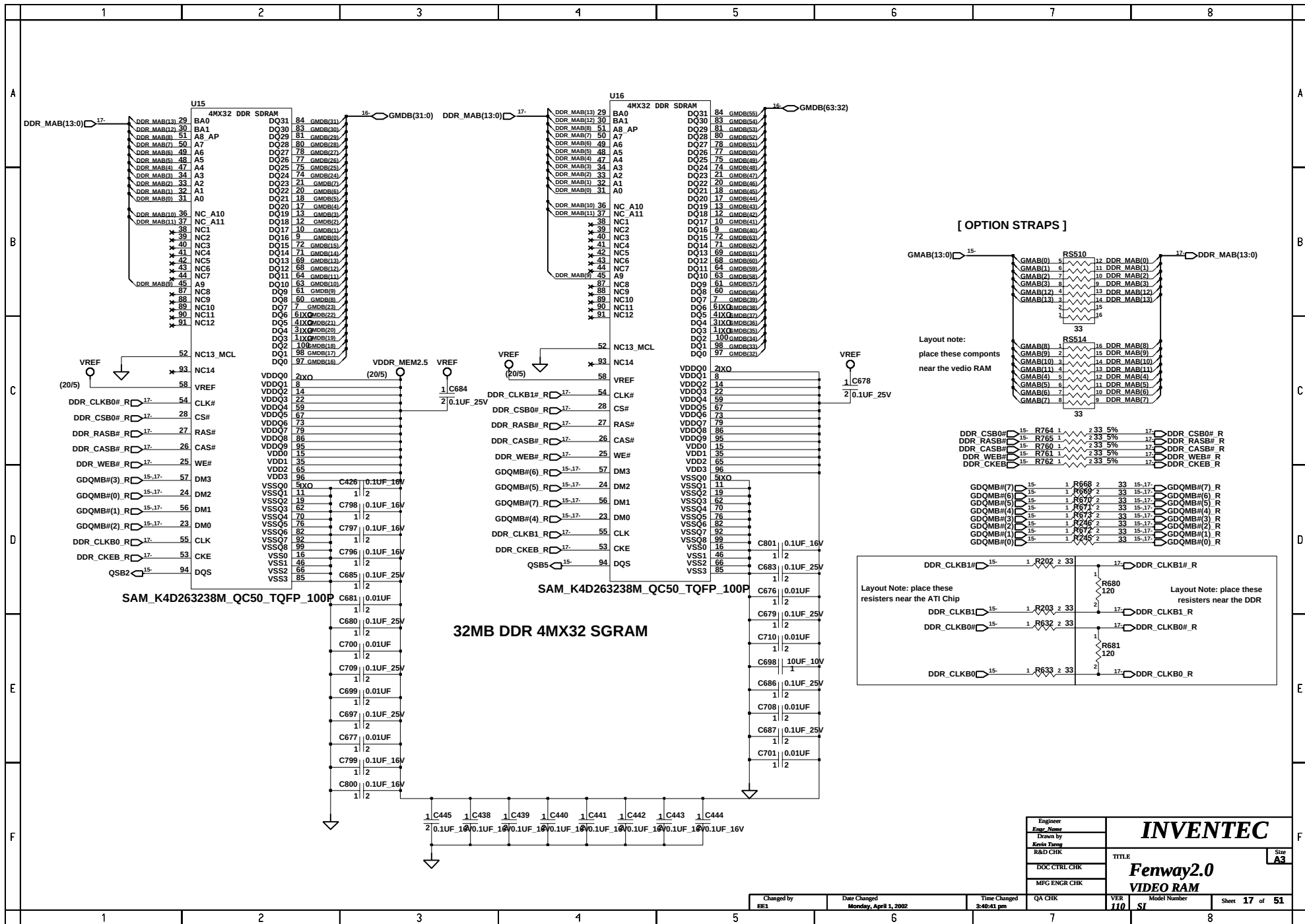
Changed by EE1	Date Changed Monday, January 28, 2002	Time Changed 11:50:49 am
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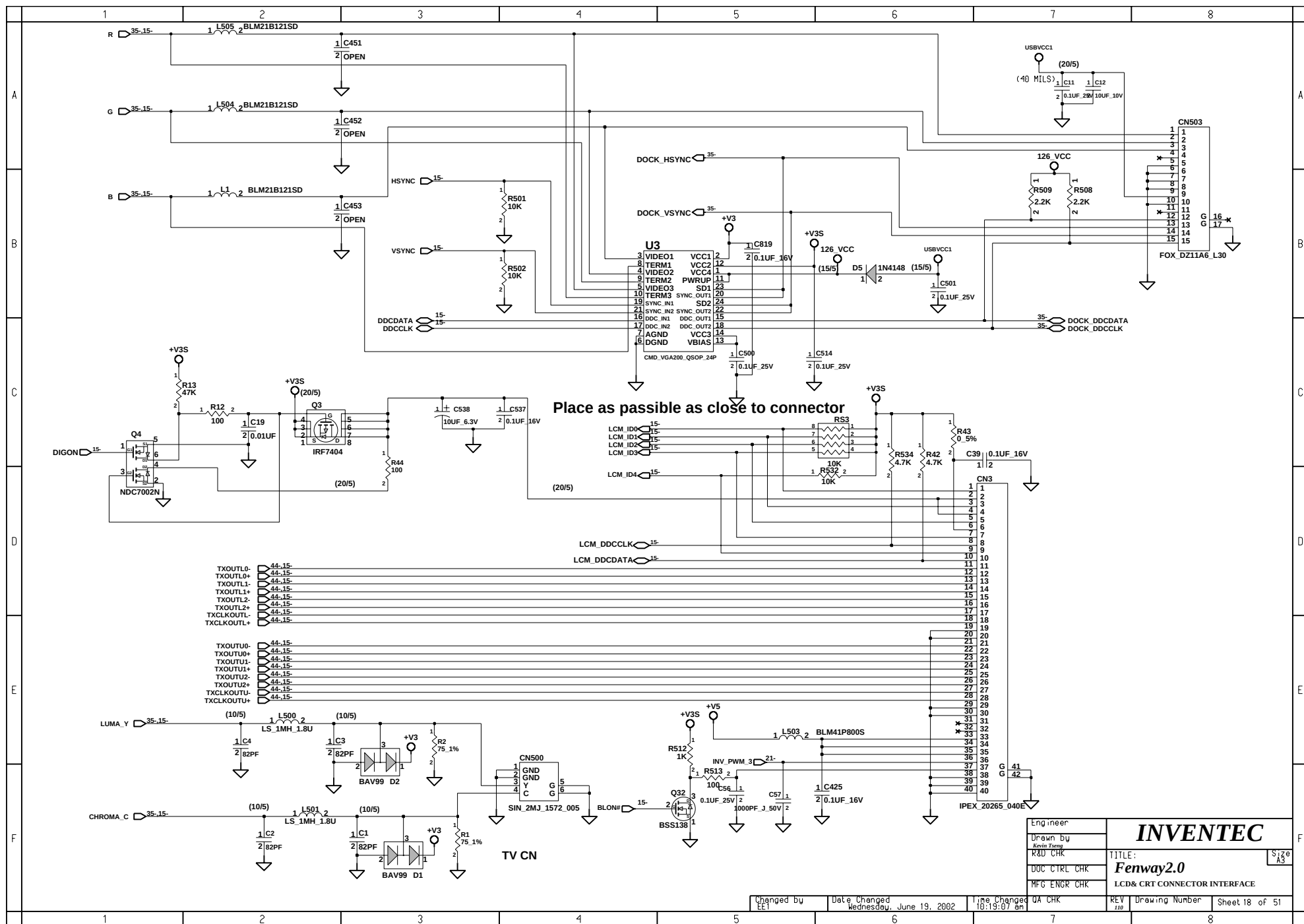


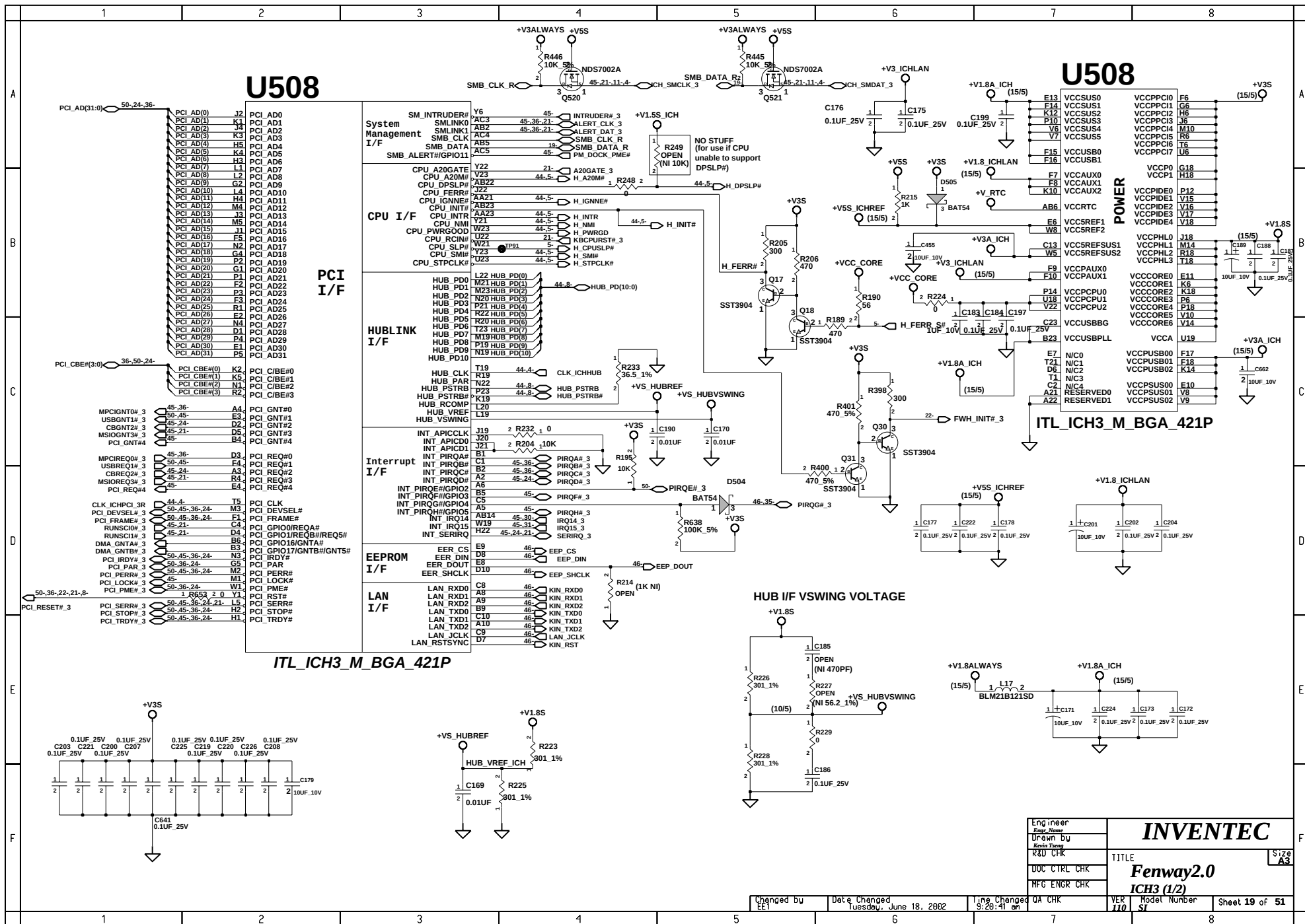
Engineer <i>Eng Name</i>	INVENTEC		
Drawn by <i>Revit Thong</i>			
R&D CHK	TITLE		S:78 A:3
DOC CTRL CHK	Fenway2.0		
MFG ENGR CHK	DDR(4/4)		
Changed by EET	Date Changed Tuesday, April 16, 2002	Time Changed 1:38:04 pm	VER 110
Model Number		Sheet 14 of 51	

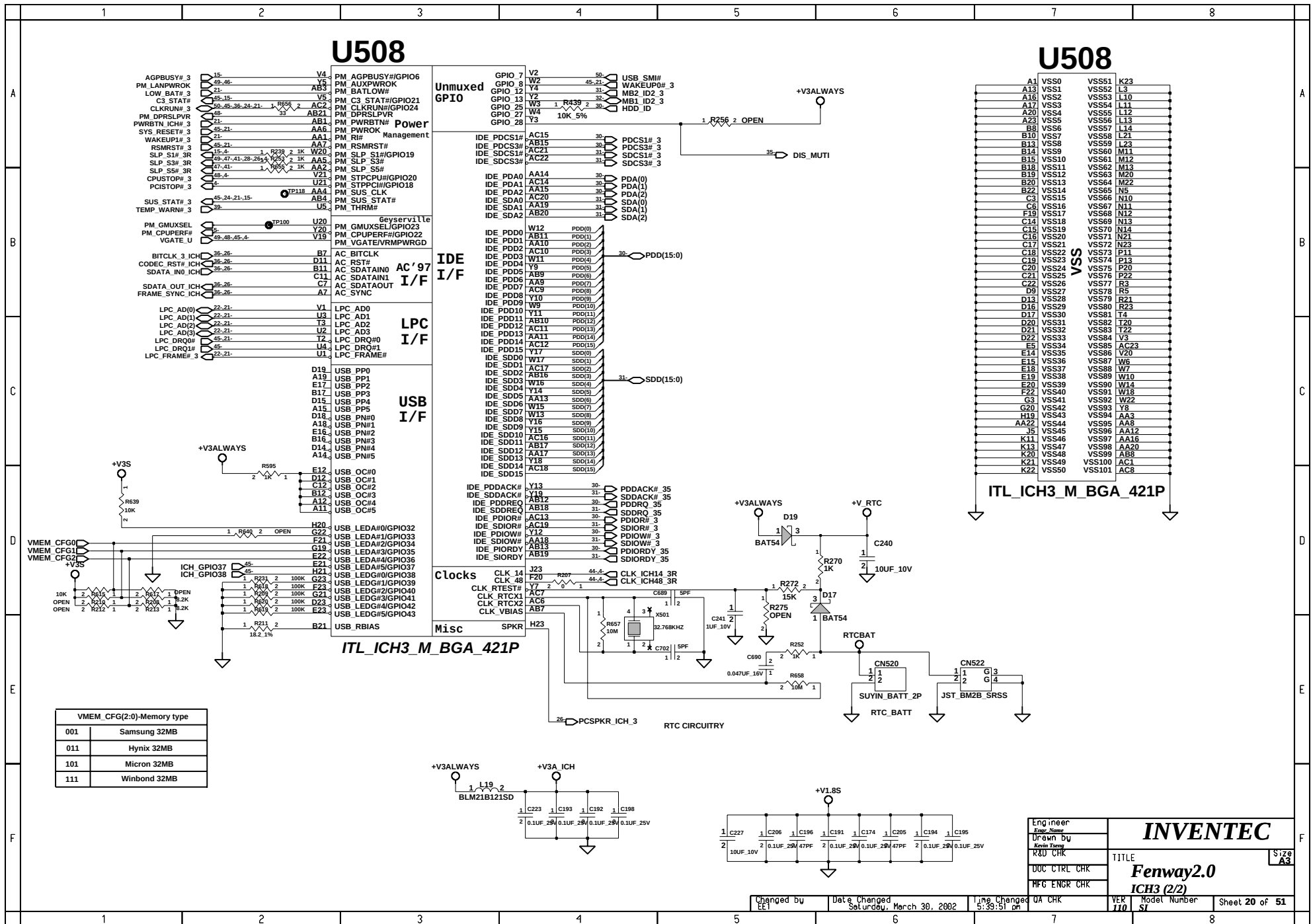




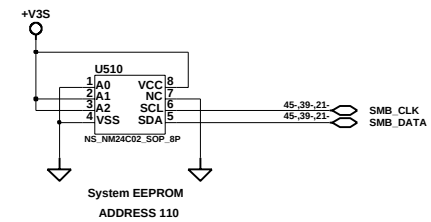
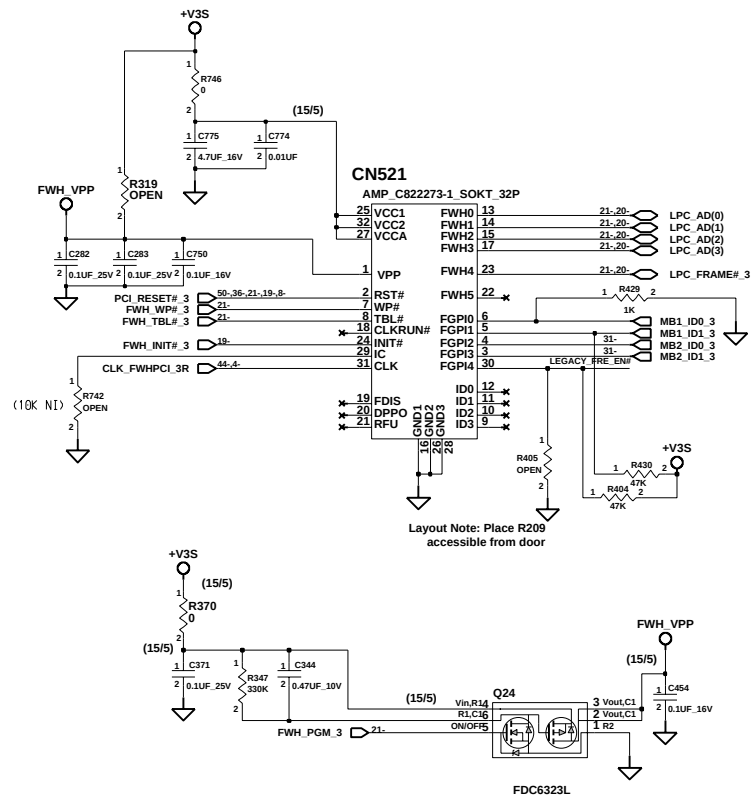






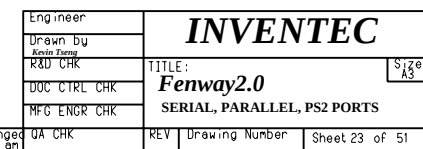


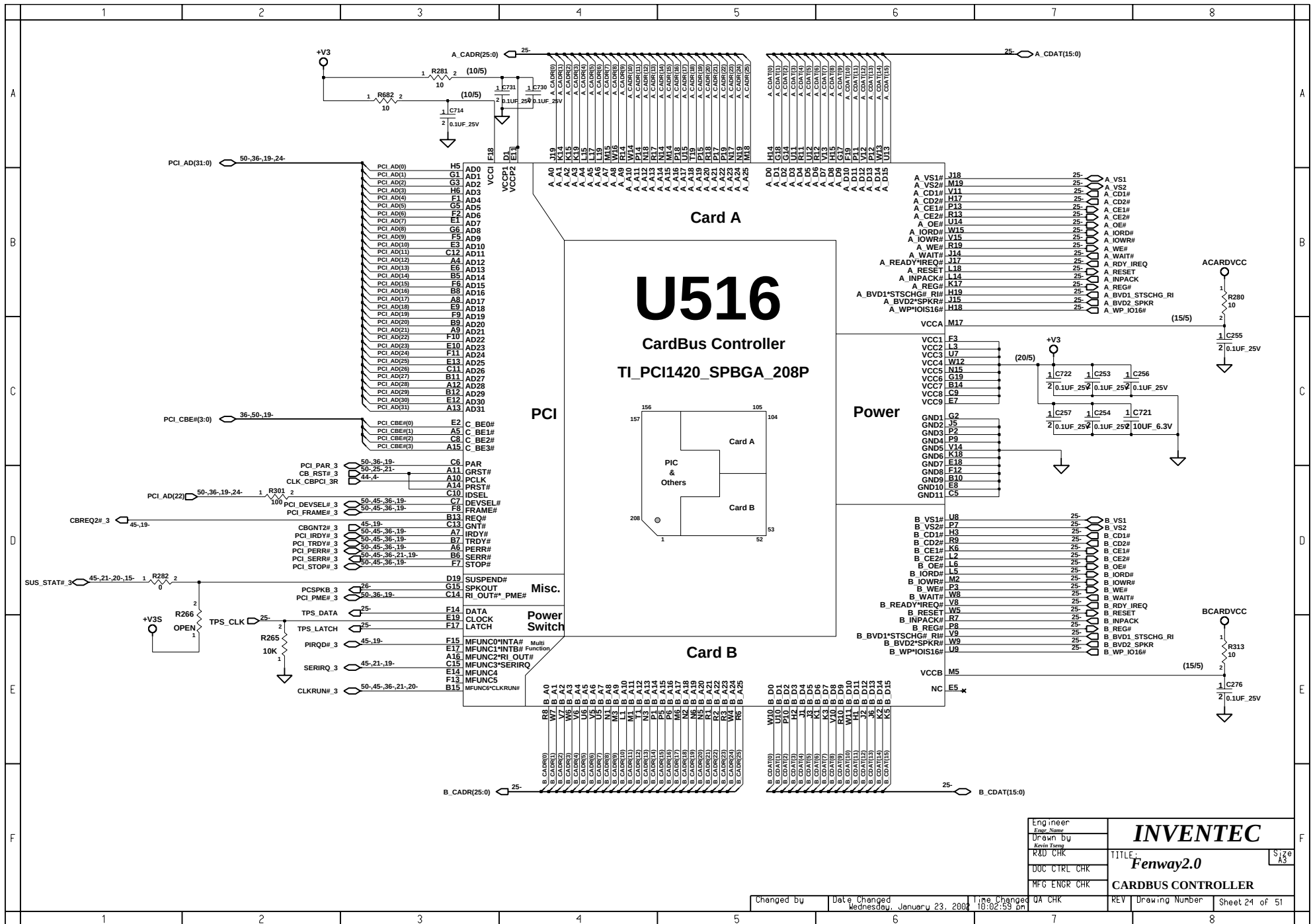
Engineer Drawn by R&D CHK	TITLE INVENTEC Fenway2.0 ICH3 (2/2)	SHEET A3
DOC CTRL CHK MFG ENGR CHK		
Changed by EE1	Date Changed Saturday, March 30, 2002	Time Changed 5:39:51 pm
VER 110	Model Number 51	Sheet 20 of 51



Engineer	INVENTEC TITLE: Fenway2.0 FWH&EEPROM		Size A3
Drawn by			
Kevin Tuong			
R&D CHK			
DOC CTRL CHK			
MFG ENGR CHK			

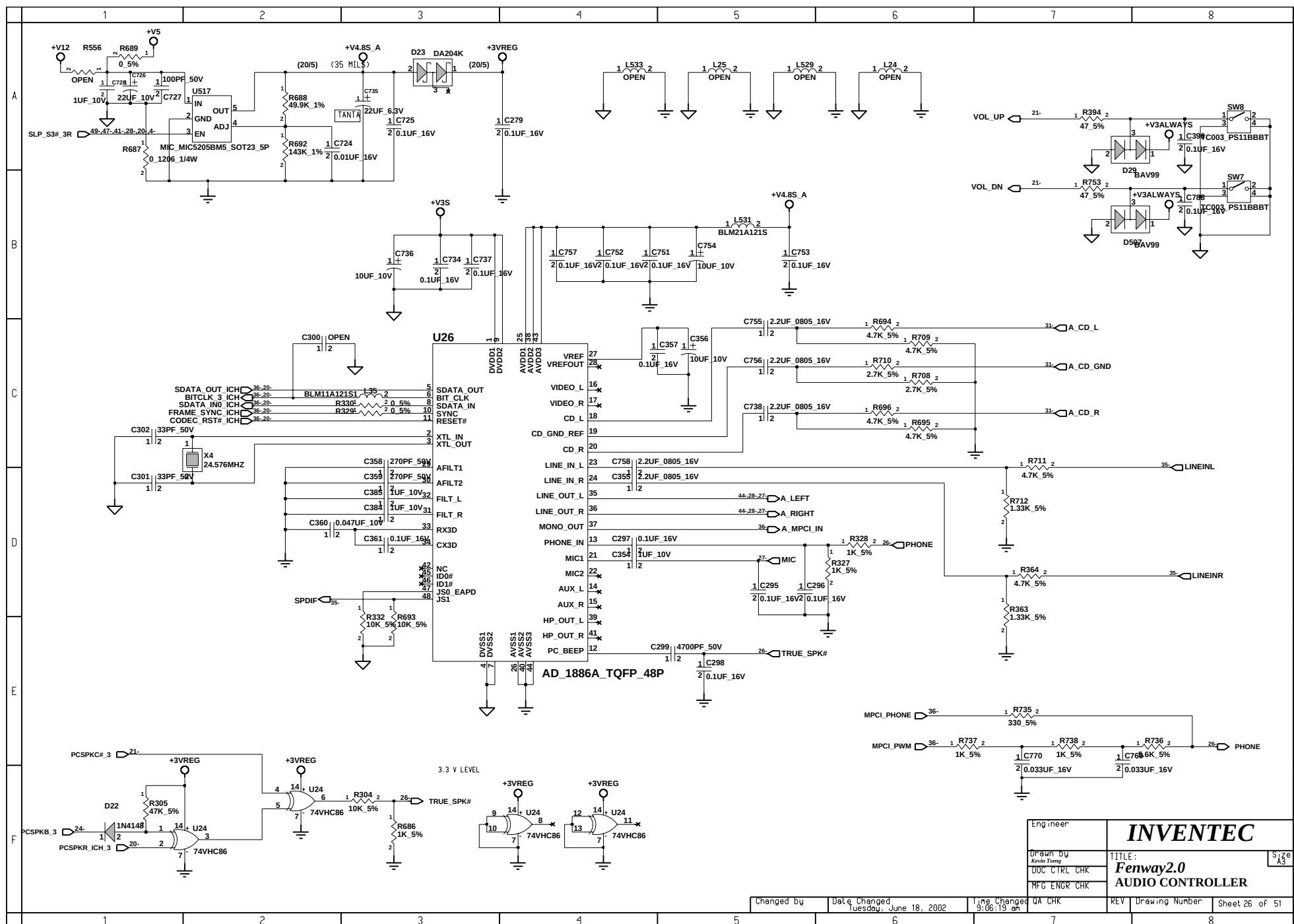
Changed by	Date Changed Tuesday, April 2, 2002	Time Changed 2:51:41 PM	QA CHK	REV	Drawing Number	Sheet 22 of 51
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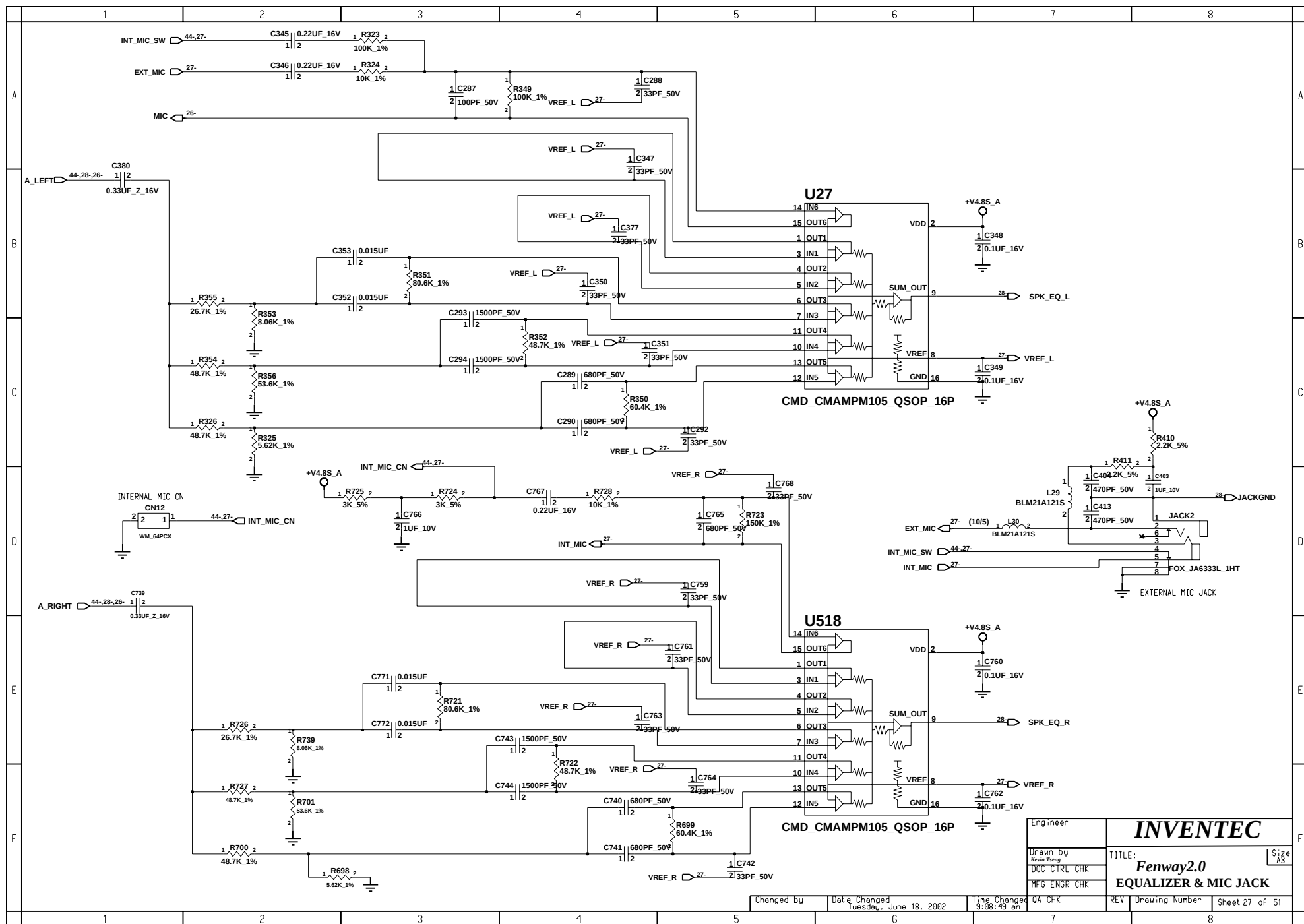


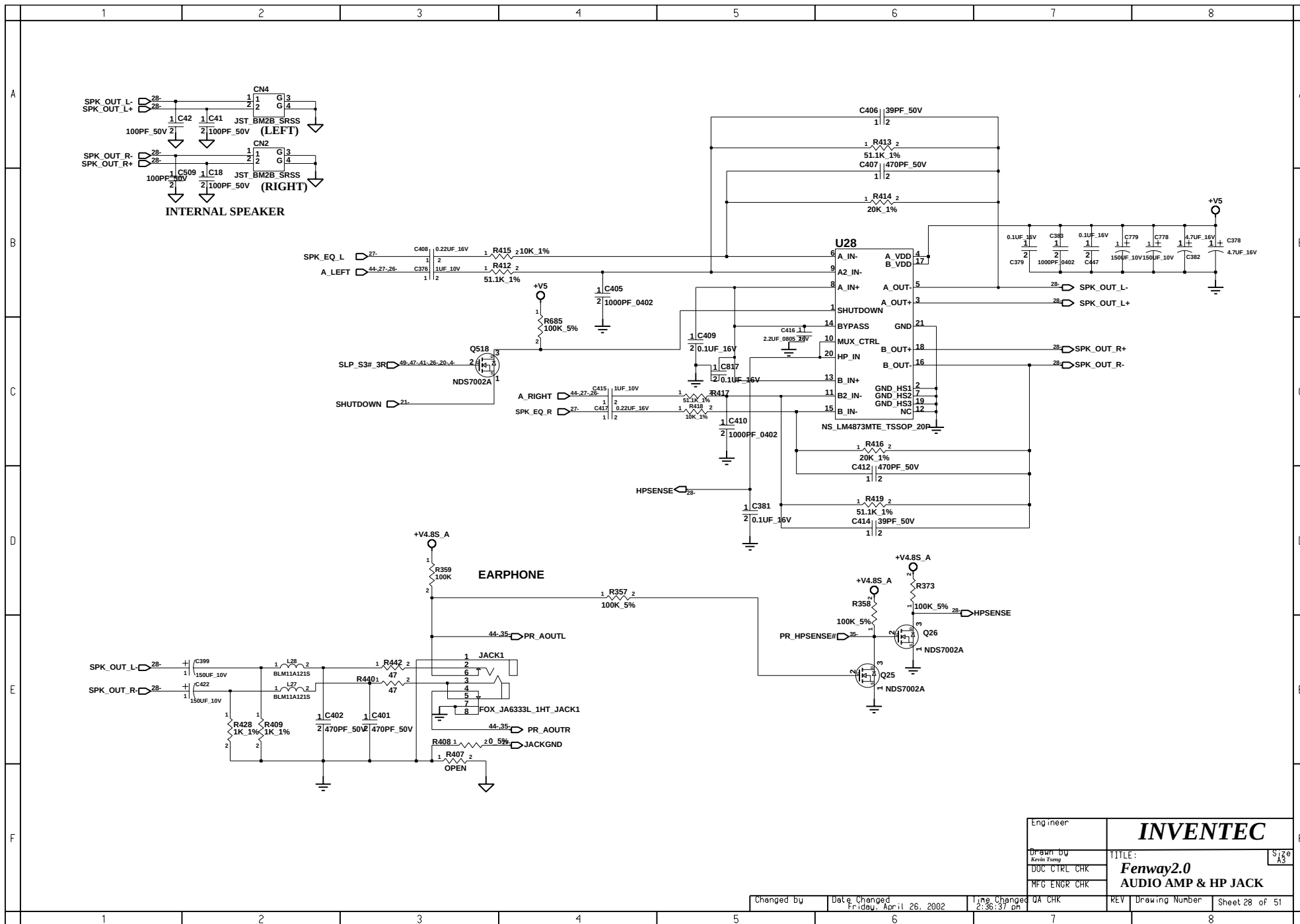


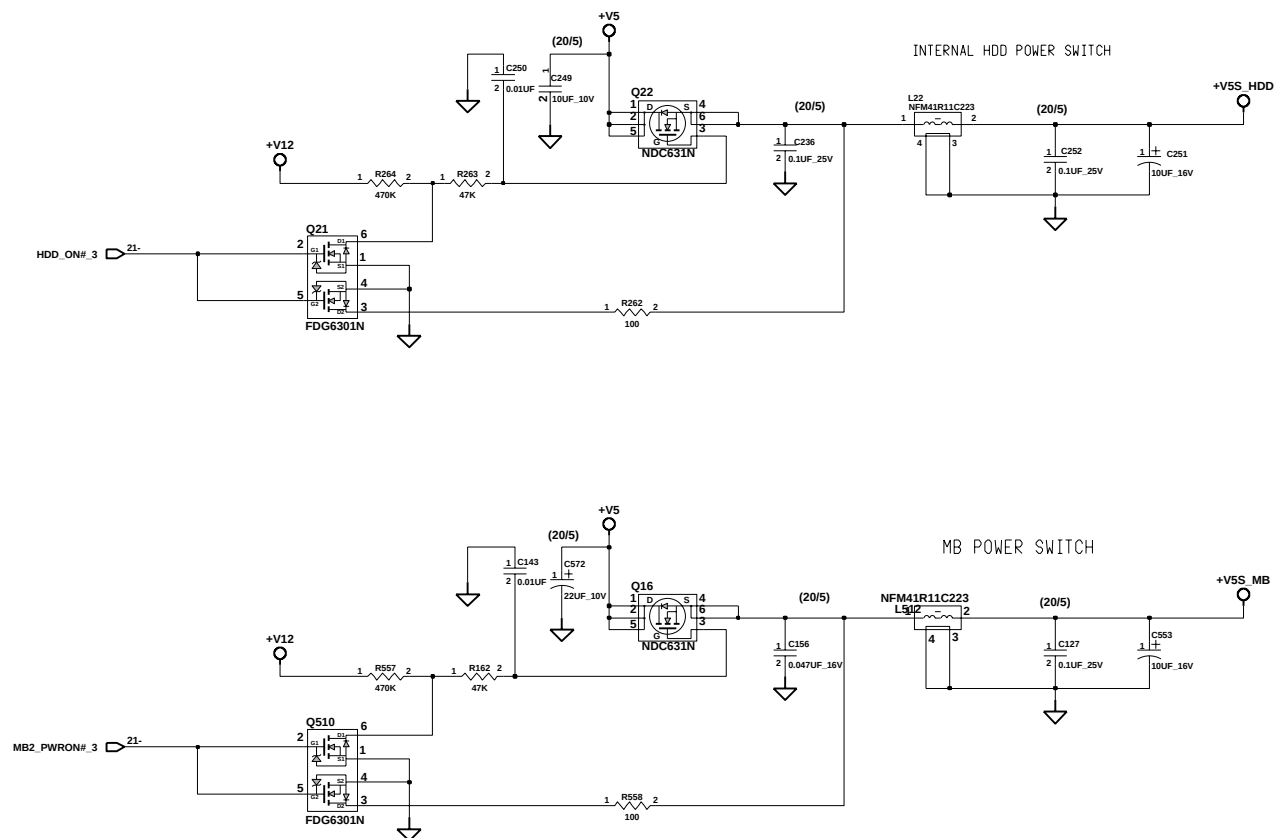
Engineer	INVENTEC
Drawn by	Fenway2.0
Rev'd	1.0
DOC CTRL CHK	CardBus Controller
MFG ENGR CHK	CardBus Controller

Changed by	Date Changed	Time Changed	Rev	Drawing Number	Sheet 24 of 51
	Wednesday, January 23, 2002	10:02:59 am	UA CHK		





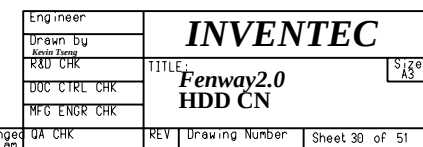




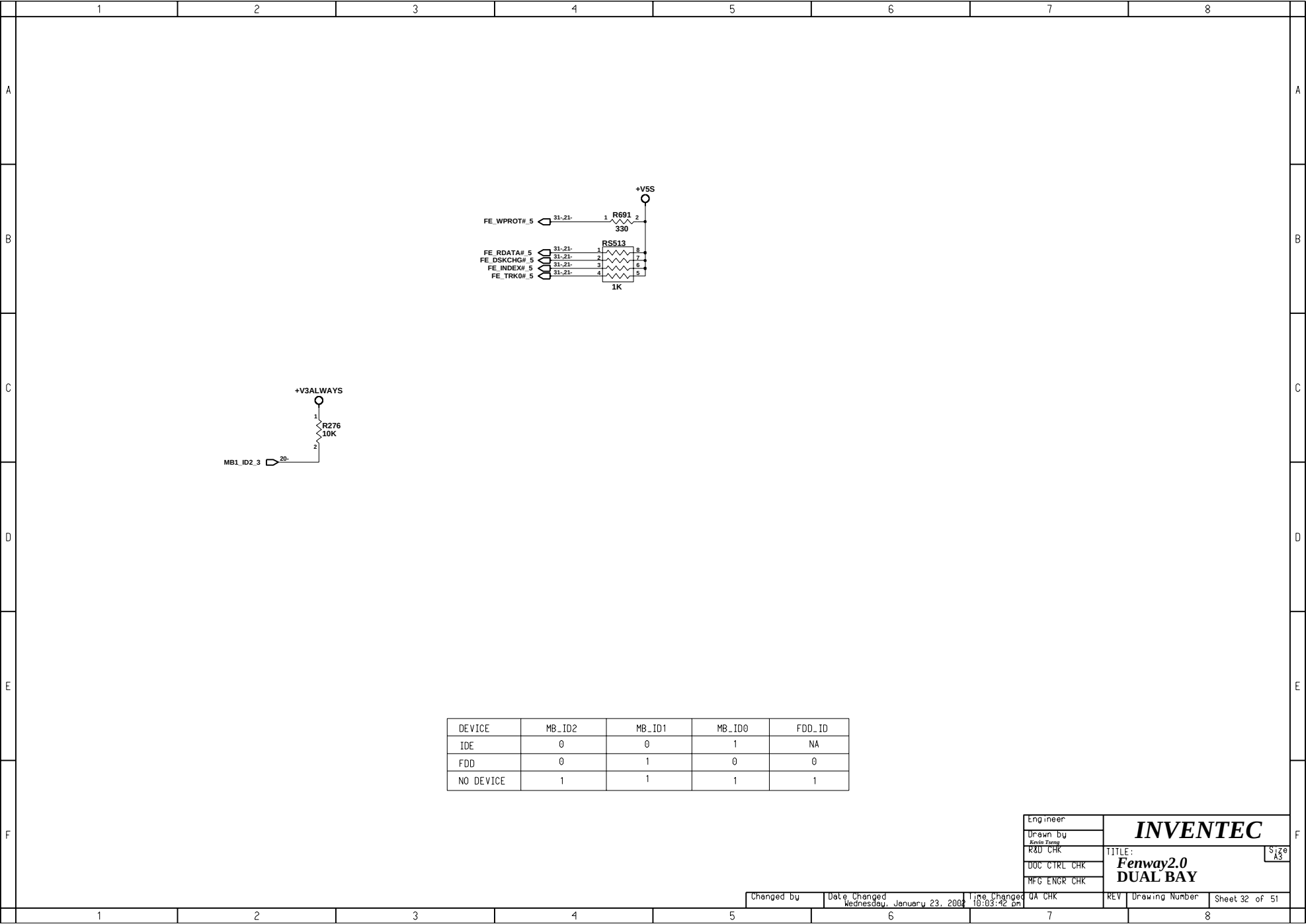
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HDDPWR	60
+V5S_MB	60
MBPWR	60

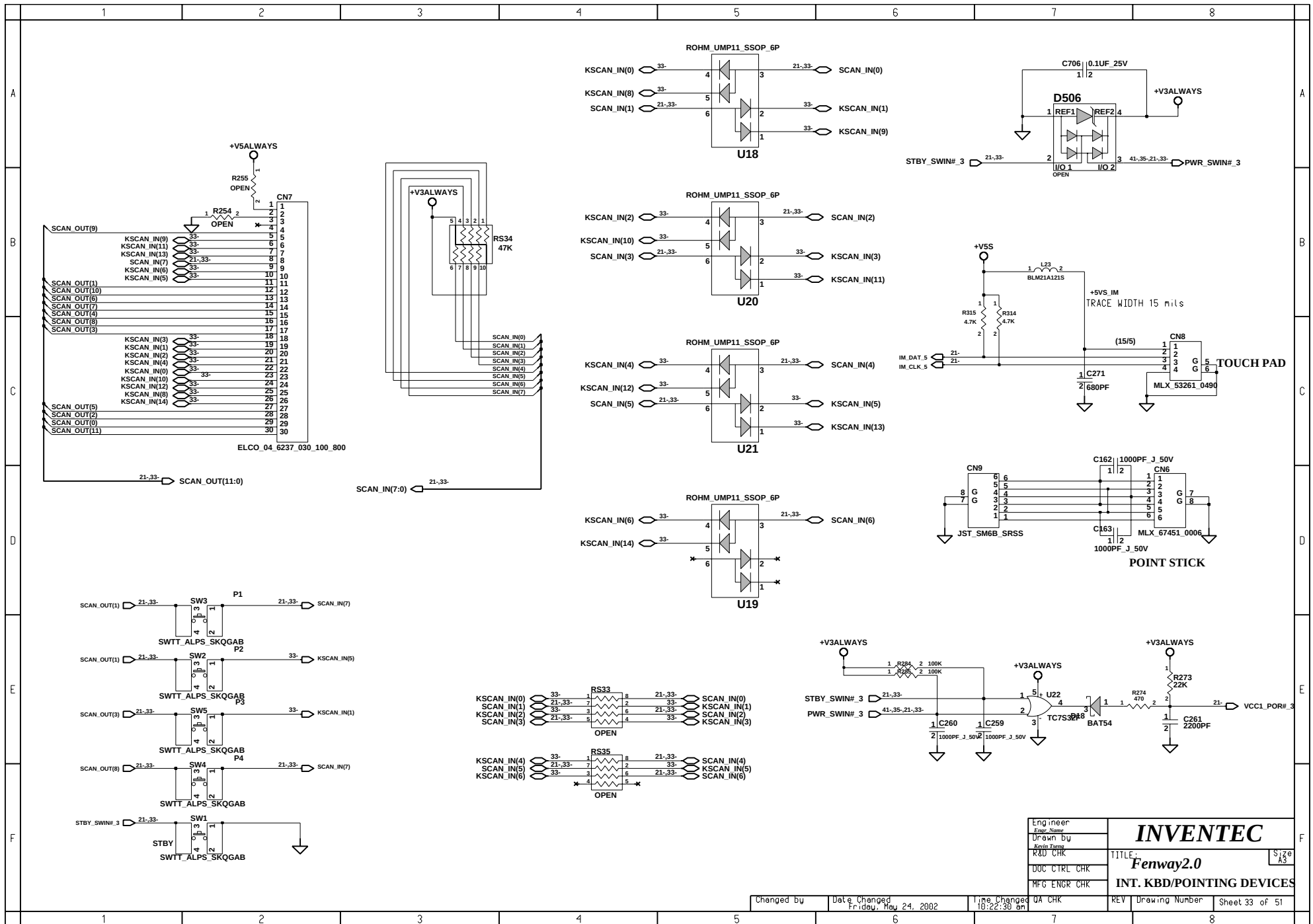
Engineer	INVENTEC TITLE: Fenway2.0 HDD & MB PWR & SWITCH	Size A3
Drawn by		
Rev'd by		
DOC CTRL CHK		
MFG ENGR CHK	REV Drawing Number Sheet 29 of 51	Time Changed 10:03:24 pm
QA CHK		
Changed by		
Date Changed		

Changed by: [Blank] Date Changed: Wednesday, January 23, 2003 Time Changed: 10:03:24 pm



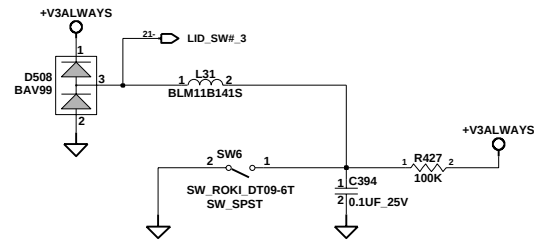




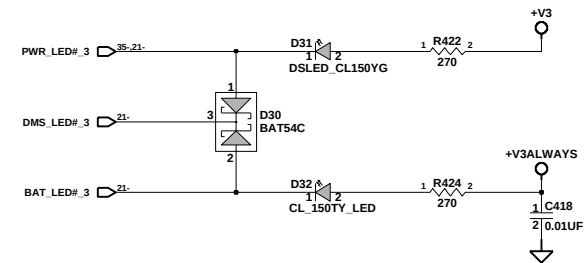
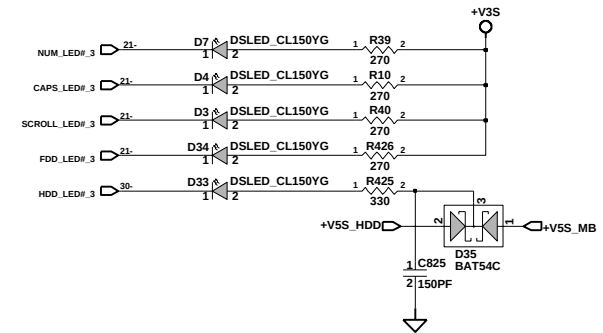


Engineer		INVENTEC TITLE: Fenway2.0 INT. KBD/POINTING DEVICES
Drawn by		
Rev'd		
DOC CTRL CHK		
MFG ENGR CHK		

Changed by	Date Changed	Time Changed	REV	Drawing Number	Sheet 33 of 51
	Friday, May 24, 2002	10:22:30 am	UA		

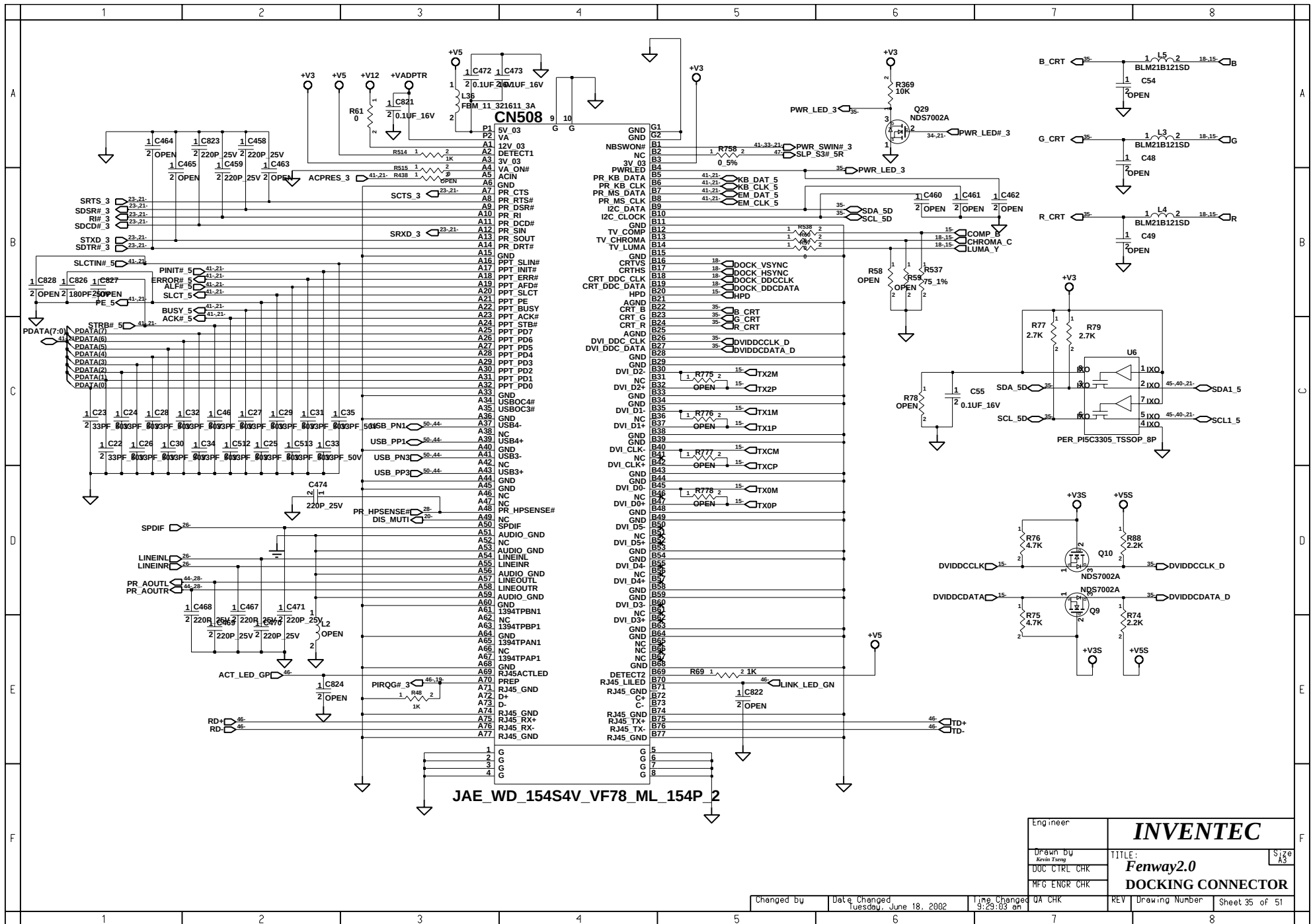


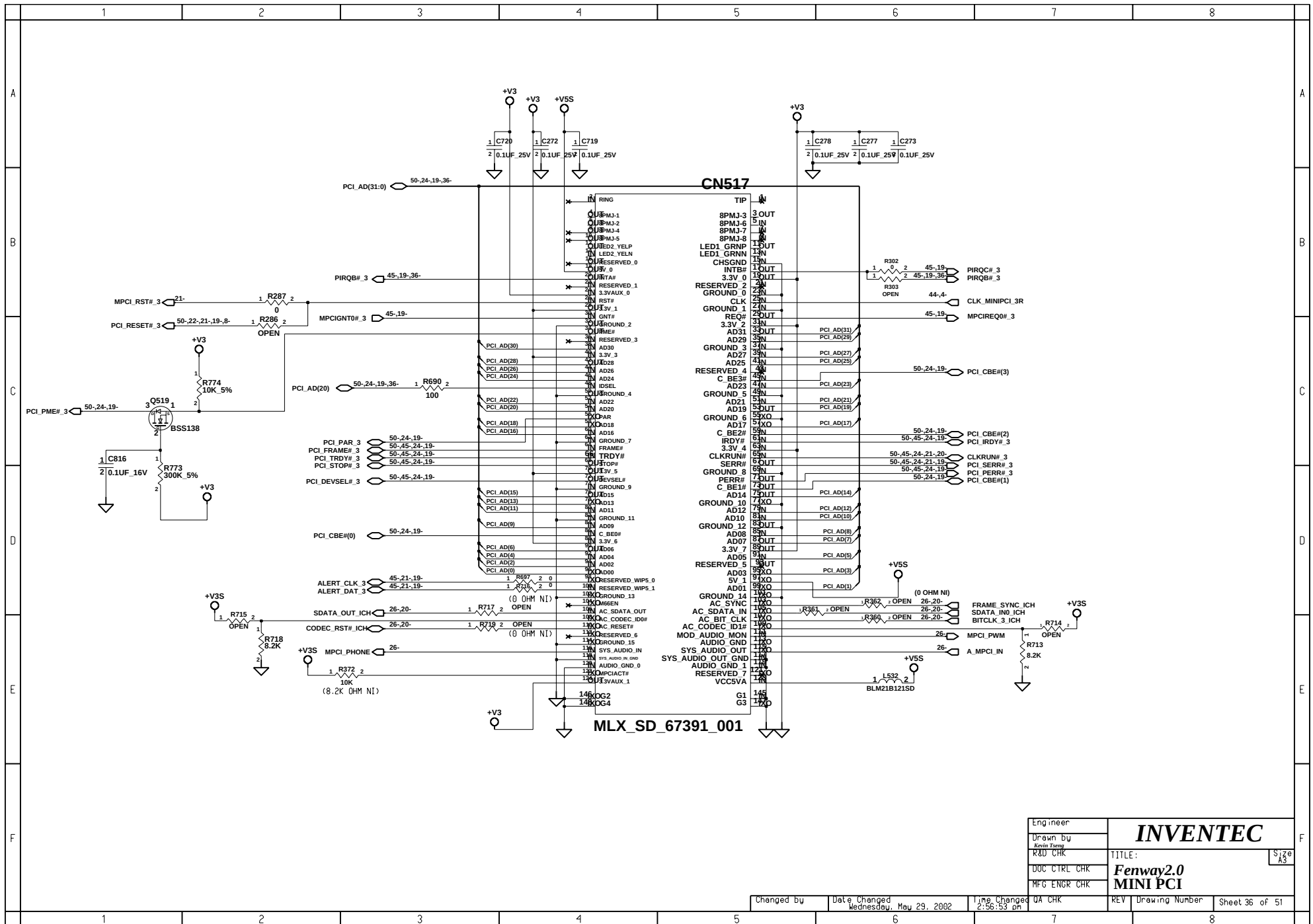
LID SWITCH



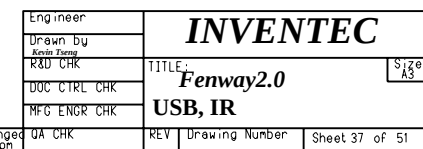
Engineer	INVENTEC	
Drawn by Kevin Tarnag	TITLE: Fenway2.0	Size A3
DOC CTRL CHK	LID SWITCHES / LEDS	
MFG ENGR CHK	REV	Drawing Number

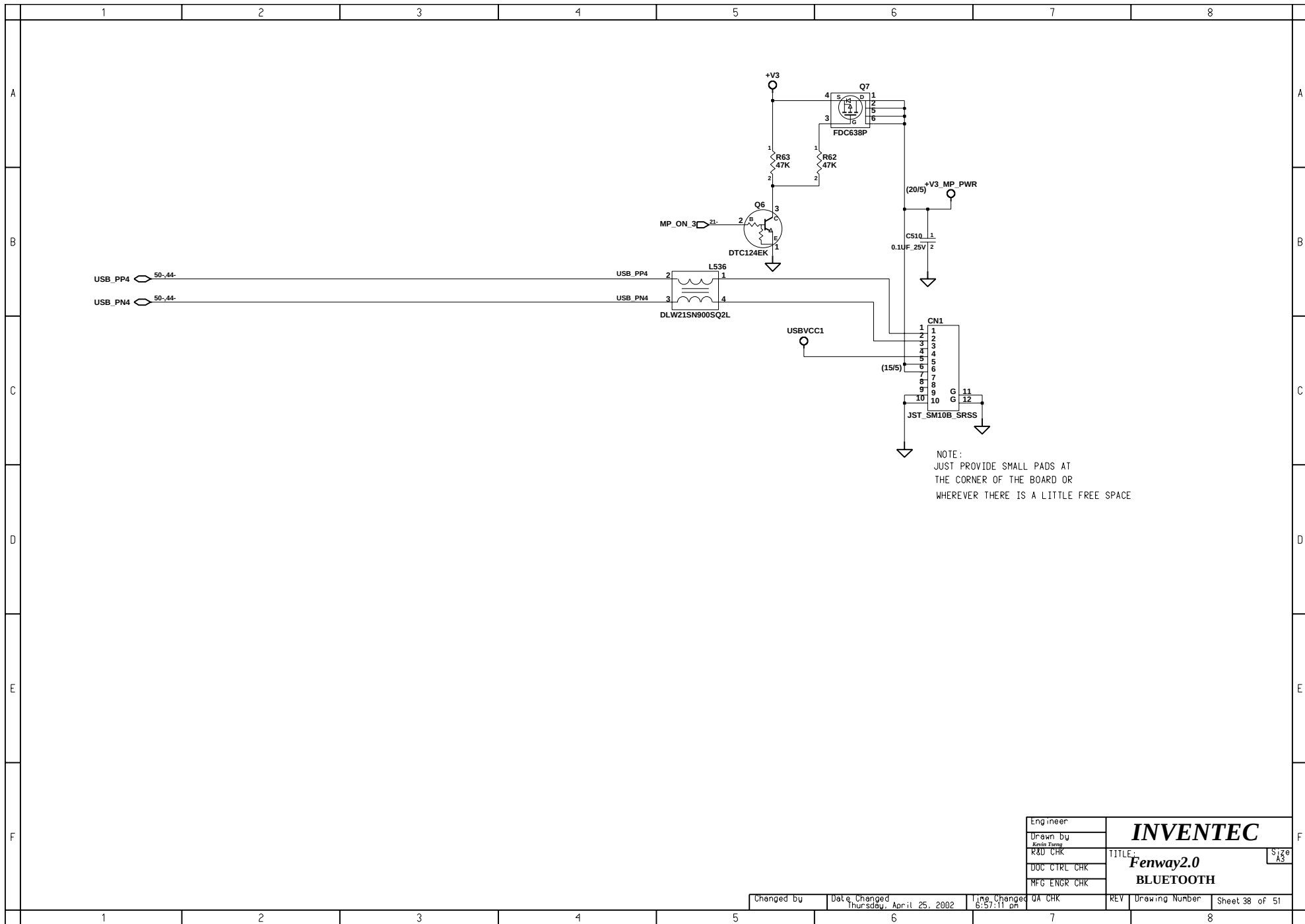
Changed by	Date Changed Thursday, May 30, 2002	Time Changed 6:33:31 pm	QA CHK	REV	Drawing Number	Sheet 34 of 51
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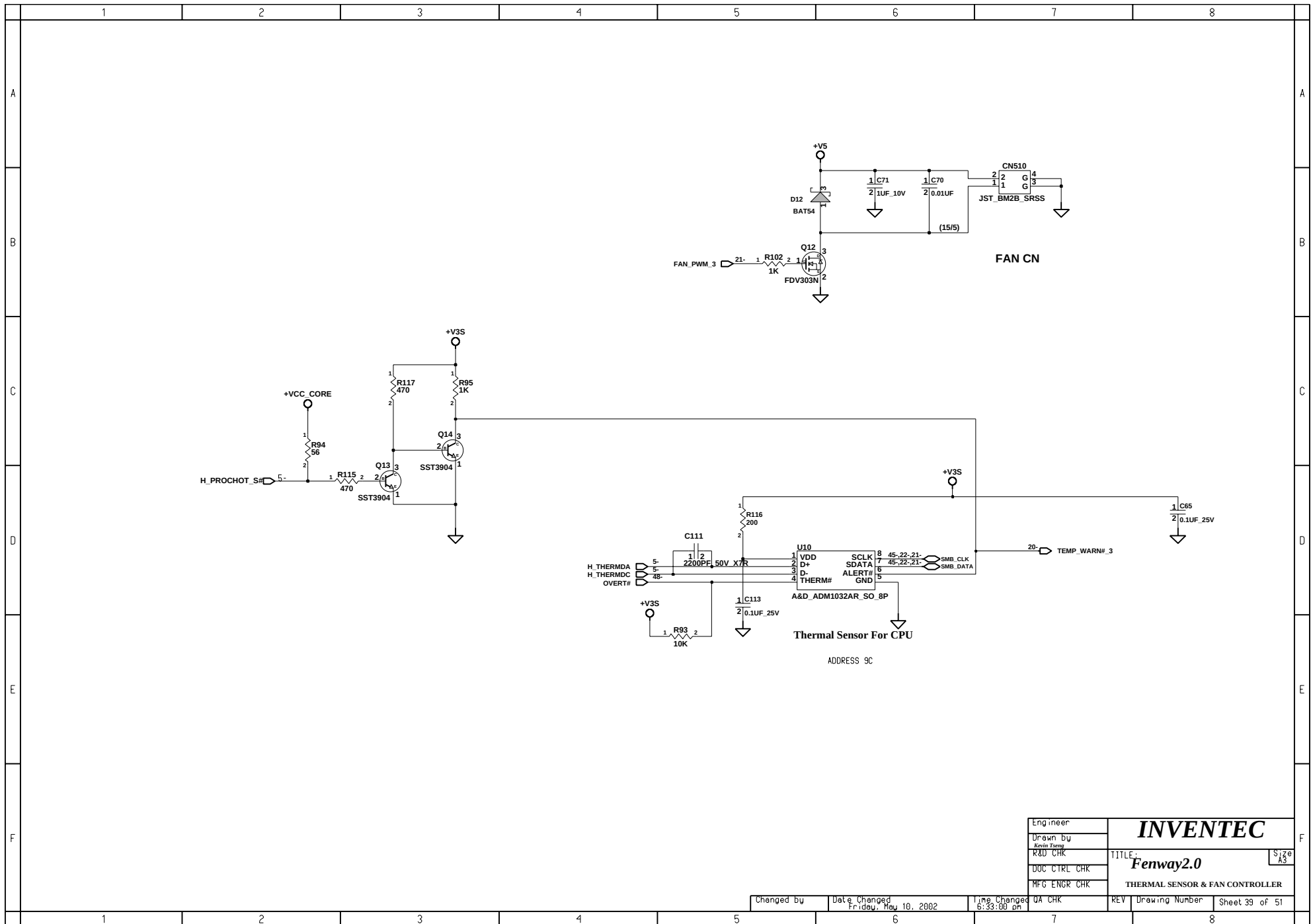


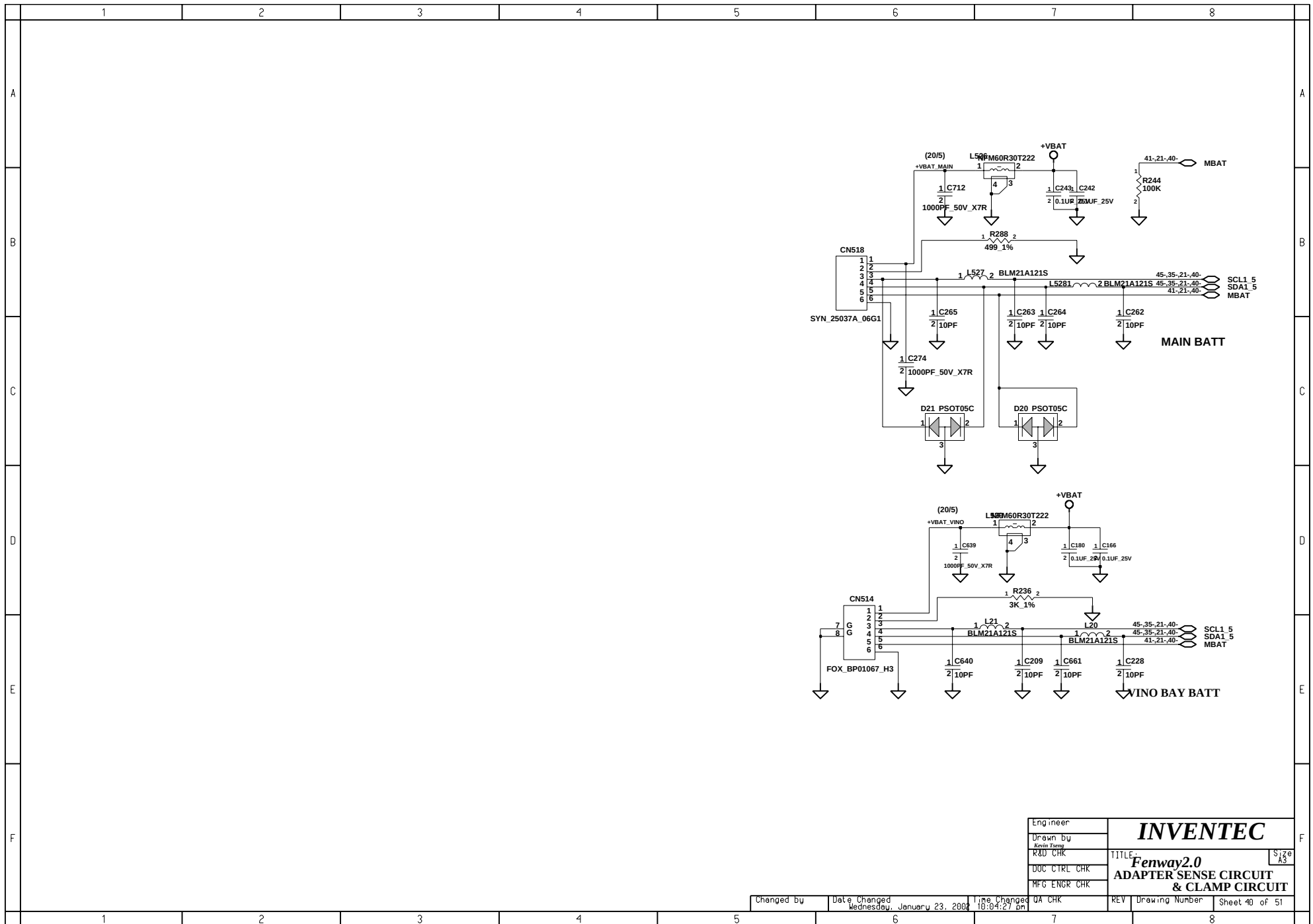


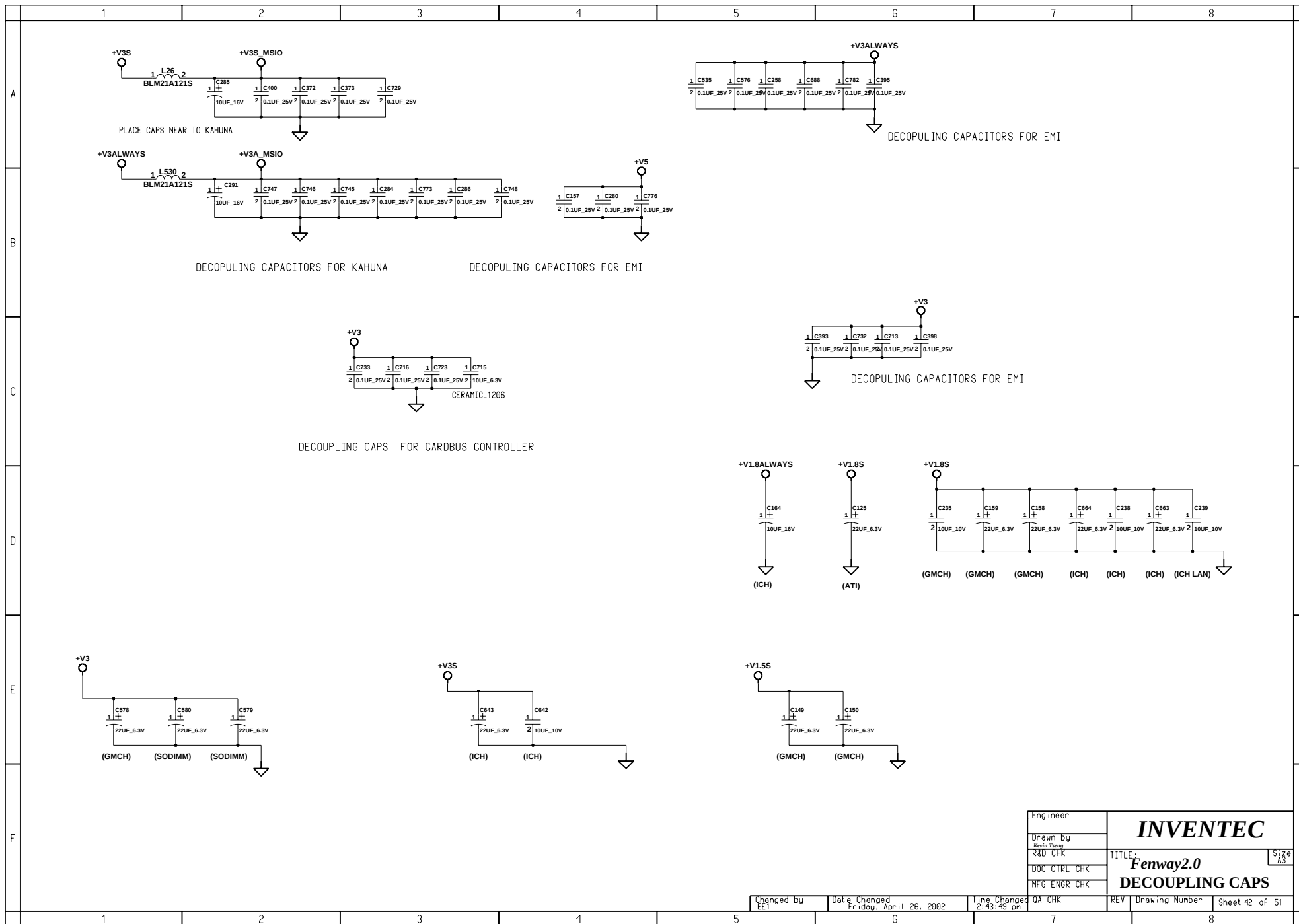
Engineer	INVENTEC		
Drawn by			
Rev'd by	TITLE: Fenway2.0		
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MFG ENGR CHK	REV: 1.0		
Changed by	Date Changed	Time Changed	Sheet 36 of 51
	Wednesday, May 29, 2002	2:56:53 pm	







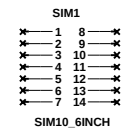




Engineer	INVENTEC TITLE: Fenway2.0 DECOUPLING CAPS	
Drawn by		
Rev'd by		
R&D CHK		
DOC CTRL CHK	Drawing Number Sheet 42 of 51	
MFG ENGR CHK		

Changed by: EET
 Date Changed: Friday, April 26, 2002
 Time Changed: 2:43:49 pm

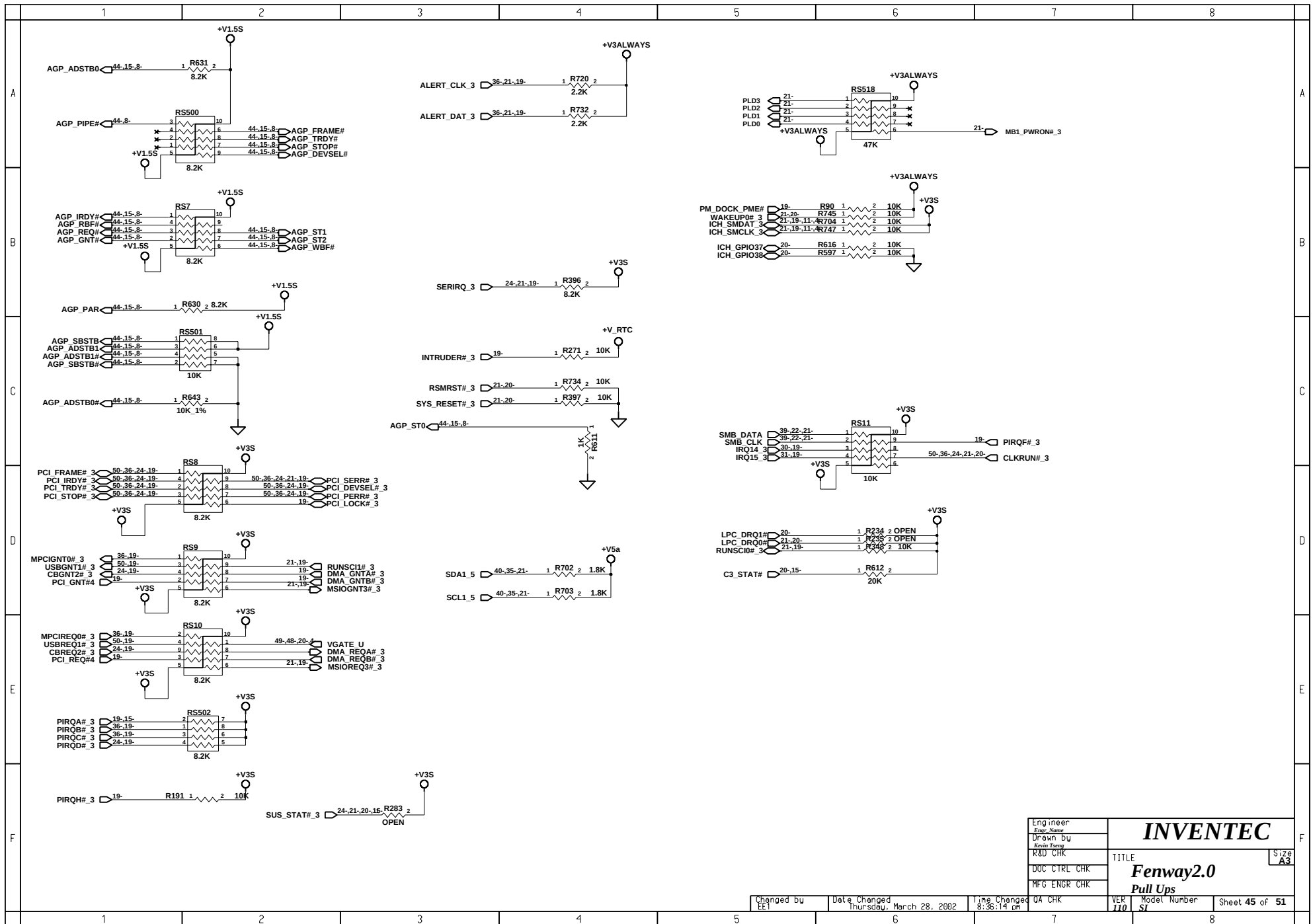
UA CHK
 REV
 Drawing Number
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FIX1 FIX3 FIX2 FIX4 FIX5 FIX6
     
 FIX_MASK FIX_MASK FIX_MASK FIX_MASK FIX_MASK FIX_MASK

Changed by	Date Changed Wednesday, January 23, 2002	Time Changed 10:04:41 am	QA CHK	REV	Drawing Number	Sheet 43 of 51
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	1	2	3	4	5	6	7	8
A	H.D#(0) 9-5 H.D#(1) 9-5 H.D#(2) 9-5 H.D#(3) 9-5 H.D#(4) 9-5 H.D#(5) 9-5 H.D#(6) 9-5 H.D#(7) 9-5 H.D#(8) 9-5 H.D#(9) 9-5 H.D#(10) 9-5 H.D#(11) 9-5 H.D#(12) 9-5 H.D#(13) 9-5 H.D#(14) 9-5 H.D#(15) 9-5 H.D#(16) 9-5 H.D#(17) 9-5 H.D#(18) 9-5 H.D#(19) 9-5 H.D#(20) 9-5 H.D#(21) 9-5 H.D#(22) 9-5 H.D#(23) 9-5 H.D#(24) 9-5 H.D#(25) 9-5 H.D#(26) 9-5 H.D#(27) 9-5 H.D#(28) 9-5 H.D#(29) 9-5 H.D#(30) 9-5 H.D#(31) 9-5 H.D#(32) 9-5 H.D#(33) 9-5 H.D#(34) 9-5 H.D#(35) 9-5 H.D#(36) 9-5 H.D#(37) 9-5 H.D#(38) 9-5 H.D#(39) 9-5 H.D#(40) 9-5 H.D#(41) 9-5 H.D#(42) 9-5 H.D#(43) 9-5 H.D#(44) 9-5 H.D#(45) 9-5 H.D#(46) 9-5 H.D#(47) 9-5 H.D#(48) 9-5 H.D#(49) 9-5 H.D#(50) 9-5 H.D#(51) 9-5 H.D#(52) 9-5 H.D#(53) 9-5 H.D#(54) 9-5 H.D#(55) 9-5 H.D#(56) 9-5 H.D#(57) 9-5 H.D#(58) 9-5 H.D#(59) 9-5 H.D#(60) 9-5 H.D#(61) 9-5 H.D#(62) 9-5 H.D#(63) 9-5	H.TRDY# 9-5 H.LOCK# 9-5 H.HITM# 9-5 H.HIT# 9-5 H.DRDY# 9-5 H.DEFER# 9-5 H.DBSY# 9-5 H.BPR# 9-5 H.BNR# 9-5 H.ADS# 9-5 H.RESET# 9-5 H.REQ#(0) 9-5 H.REQ#(1) 9-5 H.REQ#(2) 9-5 H.REQ#(3) 9-5 H.REQ#(4) 9-5 H.RS#(0) 9-5 H.RS#(1) 9-5 H.RS#(2) 9-5 H.A20M# 19-5 H.FERR# 19- H.IGNNE# 19-5 H.SMI# 9-5 H.PWRGD 19-5 H.STPKLK# 9-5 H.DPSLP# 9-5 H.INTR 19-5 H.NMI 19-5 H.INT# 19-5 H.PICD1 9-5 ITP.PREQ# 9-5 ITP.PRDY# 9-5 CLK.BCLK 9-5 CLK.BCLK R 9-5 CLK.BCLK L 9-5 CLK.BCLK R 9-5 CLK.HT 9-5 CLK.HT R 9-5 CLK.HT# 9-5 CLK.HTF R 9-5 CLK.ITP 9-5 CLK.ITP R 9-5 CLK.ITP L 9-5 CLK.ITP# R 9-5 CLK.GBOU# 15-4 CLK.AGPCONN 9-5 CLK.AGPCONN R 9-5 CLK.GBIN 9-5 CLK.GBIN R 9-5 CLK.ICHUB 9-5 CLK.ICHUB R 9-5 CLK.ICHPCI 3R 9-5 CLK.ICHPCI 3 9-5 CLK.CPU APIC 9-5 CLK.CPU APIC R 9-5 CLK.CPU APIC 2.5R 9-5 CLK.ICHAPIC 9-5 CLK.ICHAPIC R 9-5 CLK.CBPCI 3R 9-5 CLK.CBPCI 3 9-5 CLK.AUDPCI 3R 9-5 CLK.AUDPCI 3 9-5 CLK.MINIPCI 3R 9-5 CLK.MINIPCI 3 9-5 CLK.DCKPCI 3R 9-5 CLK.DCKPCI 3 9-5 CLK.FWHPCI 3R 9-5 CLK.FWHPCI 3 9-5 CLK.MSIOPCI 3R 9-5 CLK.MSIOPCI 3 9-5 CLK.ICH48 3R 9-5 CLK.ICH48 3 9-5 CLK.MSIO14 3R 9-5 CLK.MSIO14 3 9-5 CLK.ICH14 3R 9-5 CLK.ICH14 3 9-5	EXP.PCICLK 3 50-37- USB.PP0 37- USB.PP0 R 37- USB.PP0 B 37- USB.PN0 50-37- USB.PN0 R 37- USB.PN0 B 50-35- USB.PP1 50-35- EUSB.1P 3R 50-35- USB.PN1 50-35- EUSB.1N 3R 50-37- USB.PP2 37- USB.PP2 R 37- USB.PP2 B 50-37- USB.PN2 37- USB.PN2 R 50-35- USB.PP3 50-35- USB.PN3 50-38- USB.PP4 50-38- USB.PN4 R 50-38- USB.PP5 50-38- USB.PP5 3R 50-38- USB.PN5 50-38- USB.PN5 3R 50-38-	AGP.SBA(0) 15-8- AGP.SBA(1) 15-8- AGP.SBA(2) 15-8- AGP.SBA(3) 15-8- AGP.SBA(4) 15-8- AGP.SBA(5) 15-8- AGP.SBA(6) 15-8- AGP.SBA(7) 15-8- AGP.STU 45-15-8- AGP.ST1 45-15-8- AGP.ST2 45-15-8- AGP.CBE#(0) 15-8- AGP.CBE#(1) 15-8- AGP.CBE#(2) 15-8- AGP.CBE#(3) 15-8- AGP.ADI0 15-8- AGP.ADI1 15-8- AGP.ADI2 15-8- AGP.ADI3 15-8- AGP.ADI4 15-8- AGP.ADI5 15-8- AGP.ADI6 15-8- AGP.ADI7 15-8- AGP.ADI8 15-8- AGP.ADI9 15-8- AGP.ADI10 15-8- AGP.ADI11 15-8- AGP.ADI12 15-8- AGP.ADI13 15-8- AGP.ADI14 15-8- AGP.ADI15 15-8- AGP.ADI16 15-8- AGP.ADI17 15-8- AGP.ADI18 15-8- AGP.ADI19 15-8- AGP.ADI20 15-8- AGP.ADI21 15-8- AGP.ADI22 15-8- AGP.ADI23 15-8- AGP.ADI24 15-8- AGP.ADI25 15-8- AGP.ADI26 15-8- AGP.ADI27 15-8- AGP.ADI28 15-8- AGP.ADI29 15-8- AGP.ADI30 15-8- AGP.ADI31 15-8- AGP.PIPE# 45-8- AGP.RBF# 45-15-8- AGP.WBF# 45-15-8- AGP.FRAME# 45-15-8- AGP.IRDY# 45-15-8- AGP.TRDY# 45-15-8- AGP.STOP# 45-15-8- AGP.DEVSEL# 45-15-8- AGP.REQ# 45-15-8- AGP.GNT# 45-15-8- AGP.PAR 45-15-8- AGP.ADSTB0 45-15-8- AGP.ADSTB0W 45-15-8- AGP.ADSTB1 45-15-8- AGP.ADSTB1W 45-15-8- AGP.SBSTB 45-15-8- AGP.SBSTB# 45-15-8-	HUB.PSTRB# 19-8- HUB.PD(0) 19-8- HUB.PD(1) 19-8- HUB.PD(2) 19-8- HUB.PD(3) 19-8- HUB.PD(4) 19-8- HUB.PD(5) 19-8- HUB.PD(6) 19-8- HUB.PD(7) 19-8- HUB.PD(8) 19-8- HUB.PD(9) 19-8- HUB.PD(10) 19-8- A.LEFT 28-27-26- A.RIGHT 28-27-26- PR.AOUT# 35-28- PR.AOUT 35-28- INT.MIC.SW 27- INT.MIC.CN 27- TXOUTL0 18-15- TXOUTL0 18-15- TXOUTL1 18-15- TXOUTL1 18-15- TXOUTL2 18-15- TXOUTL2 18-15- TXCLKOUTL 18-15- TXCLKOUTL 18-15- TXOUTU0 18-15- TXOUTU0 18-15- TXOUTU1 18-15- TXOUTU1 18-15- TXOUTU2 18-15- TXOUTU2 18-15- TXCLKOUTU 18-15- TXCLKOUTU 18-15-			
B								
C								
D	H.A#(3) 9-5 H.A#(4) 9-5 H.A#(5) 9-5 H.A#(6) 9-5 H.A#(7) 9-5 H.A#(8) 9-5 H.A#(9) 9-5 H.A#(10) 9-5 H.A#(11) 9-5 H.A#(12) 9-5 H.A#(13) 9-5 H.A#(14) 9-5 H.A#(15) 9-5 H.A#(16) 9-5 H.A#(17) 9-5 H.A#(18) 9-5 H.A#(19) 9-5 H.A#(20) 9-5 H.A#(21) 9-5 H.A#(22) 9-5 H.A#(23) 9-5 H.A#(24) 9-5 H.A#(25) 9-5 H.A#(26) 9-5 H.A#(27) 9-5 H.A#(28) 9-5 H.A#(29) 9-5 H.A#(30) 9-5 H.A#(31) 9-5							
E								
F								
					</			



Engineer	INVENTEC TITLE Fenway2.0 Pull Ups		
Drawn by			
Rev'd			
DOC CTRL			
MFG ENGR			
Changed by	Date Changed	Time Changed	VER
EE1	Thursday, March 28, 2002	8:36:14 pm	110
Model Number	Sheet 45 of 51		
51			

Layout notes:
Use 50 ohm traces.

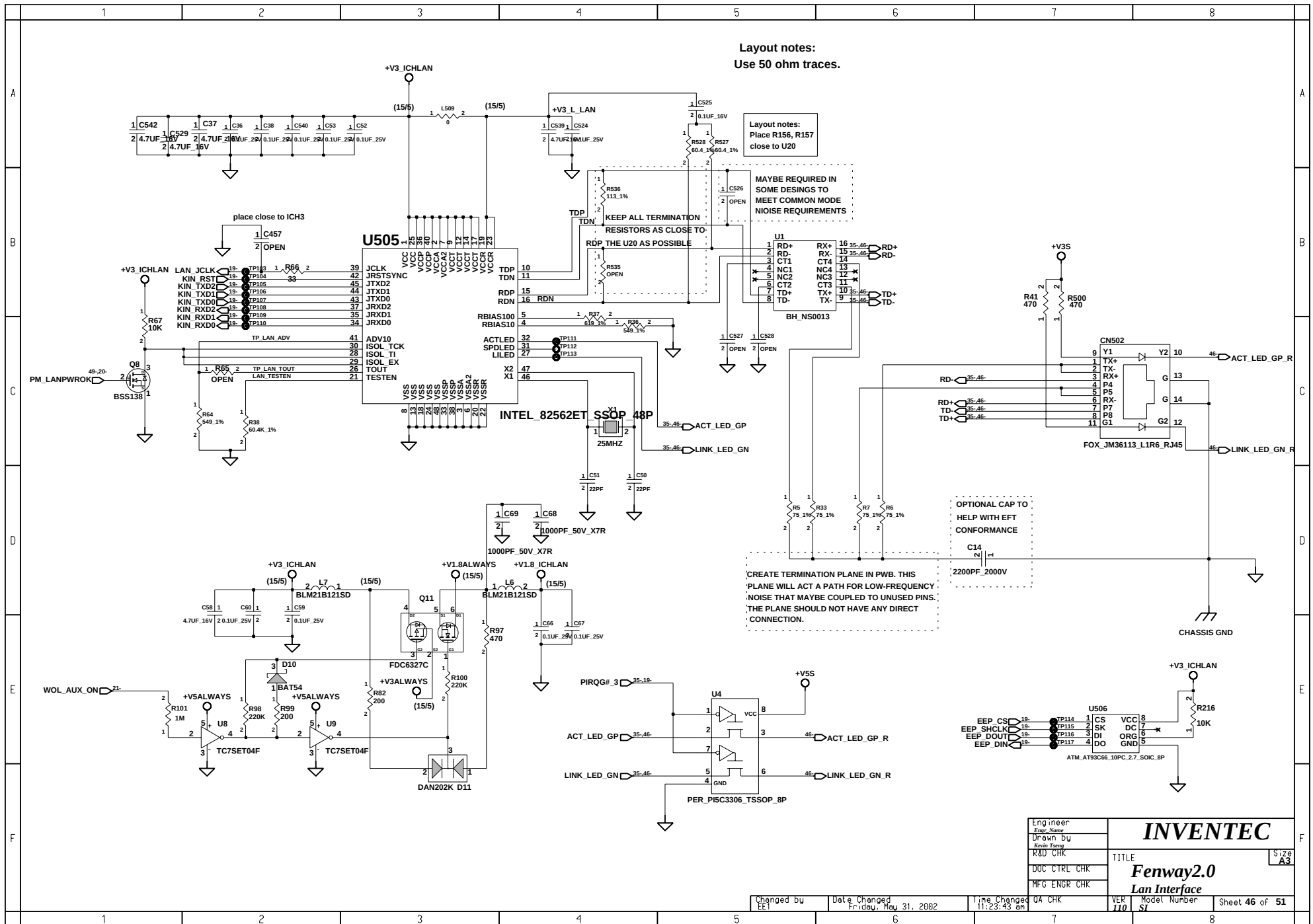
Layout notes:
Place R156, R157
close to U20

MAYBE REQUIRED IN
SOME DESIGNS TO
MEET COMMON MODE
NOISE REQUIREMENTS

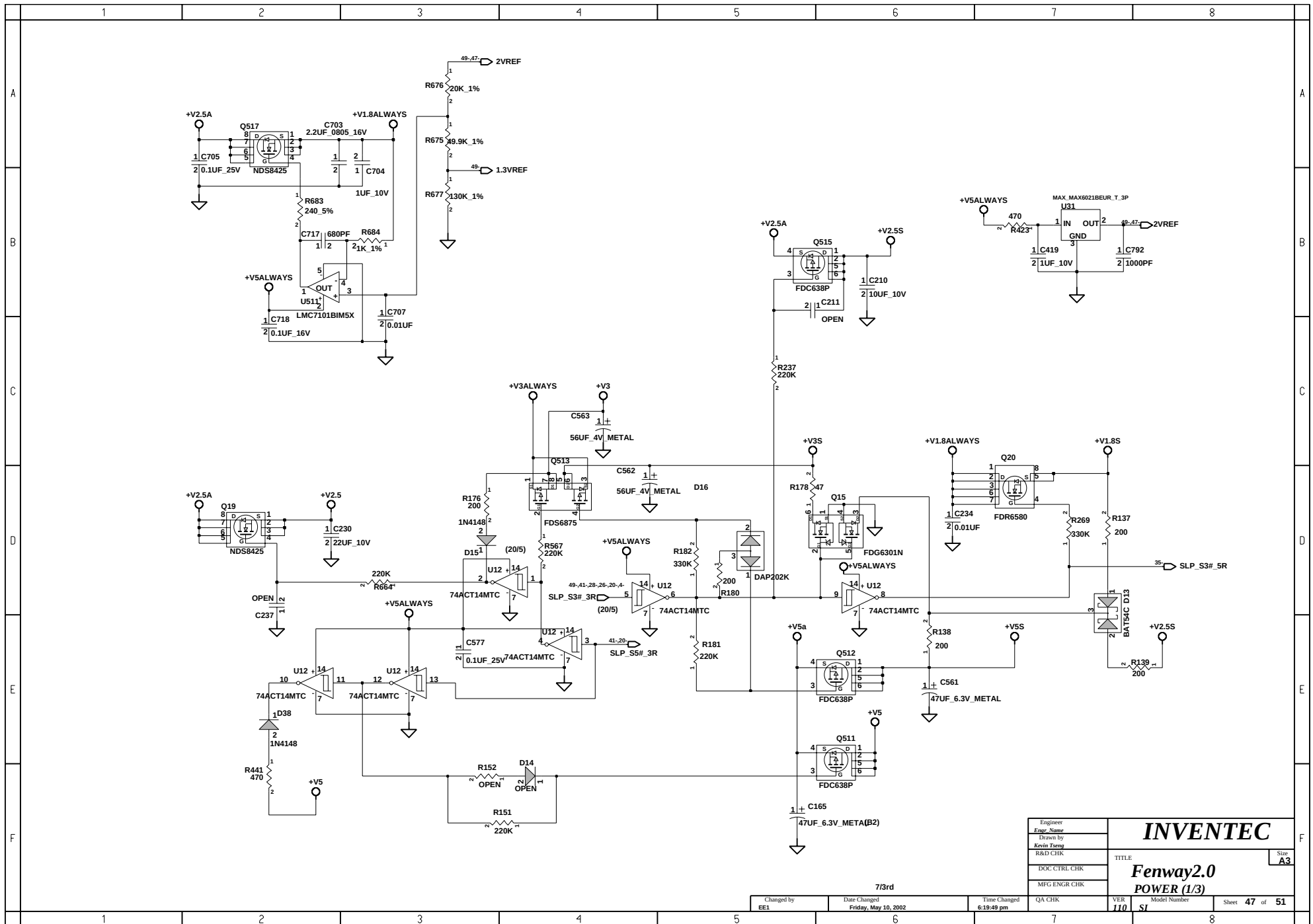
KEEP ALL TERMINATION
RESISTORS AS CLOSE TO
THE U20 AS POSSIBLE

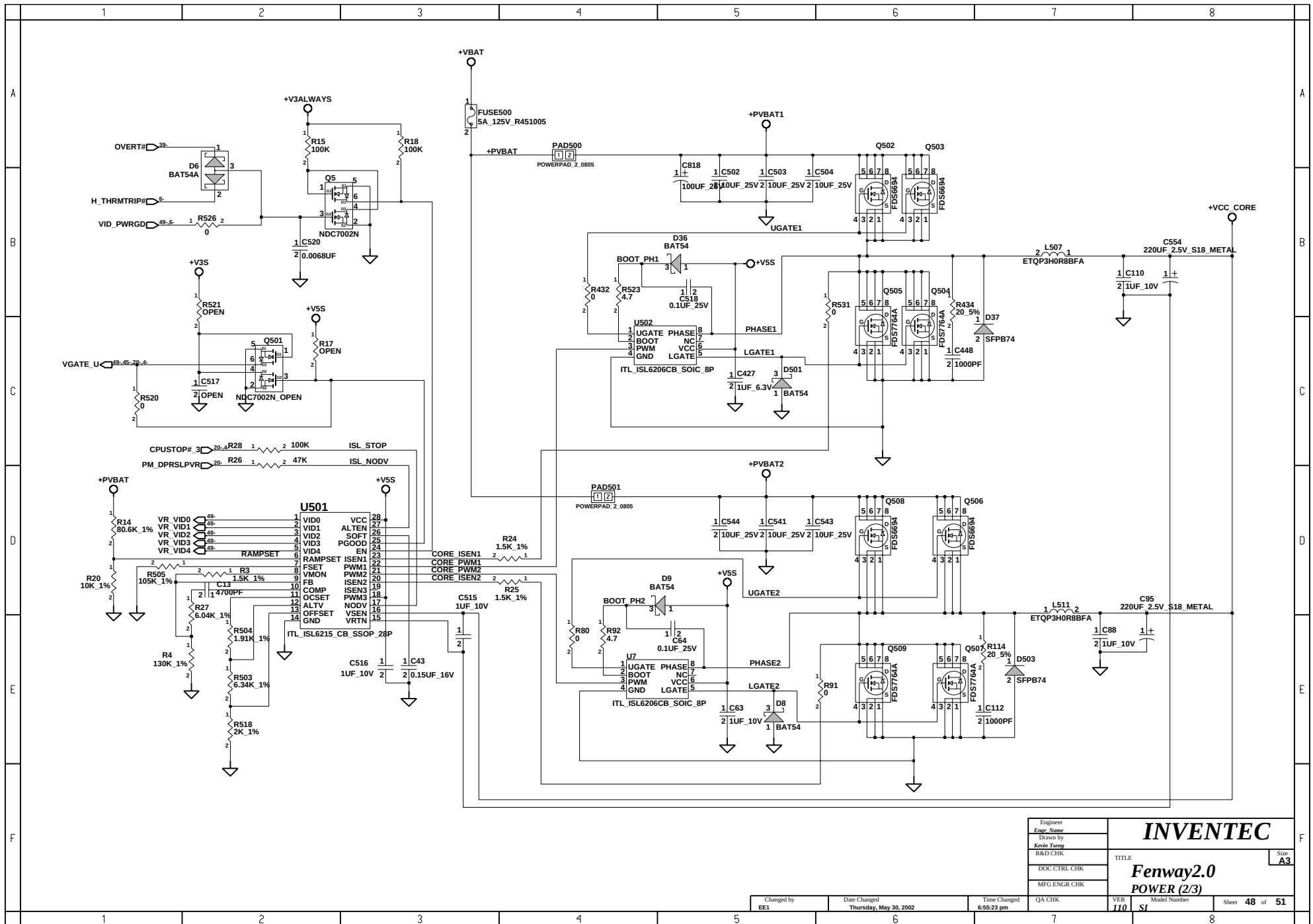
OPTIONAL CAP TO
HELP WITH EFT
CONFORMANCE

CREATE TERMINATION PLANE IN PWB. THIS
PLANE WILL ACT A PATH FOR LOW-FREQUENCY
NOISE THAT MAYBE COUPLED TO UNUSED PINS.
THE PLANE SHOULD NOT HAVE ANY DIRECT
CONNECTION.



Engineer		INVENTEC TITLE Fenway2.0 Lan Interface Model Number 110	S:78 A:3 Sheet 46 of 51
Drawn by			
Rev'd by			
DOC CTRL CHK			
MFG ENGR CHK			
Changed by	EE1	Time Changed	11:23:43 am
Date Changed	Friday, May 31, 2002	Model Number	110





Engineer	
Eng. Name	
Drawn by	
Rev'd	
R&D CHK	
DOC CTRL CHK	
MFG ENGR CHK	
VER	110
Model Number	SI
Sheet	48 of 51

INVENTEC

TITLE: **Fenway2.0**

POWER (2/3)

Changed by	Date Changed	Time Changed	QA CHK
EE1	Thursday, May 30, 2002	6:55:23 pm	

